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Volume 01 — Networking Foundations (L01-L10)

Course: NPTEL Ethical Hacking (106105217) · **Instructor:** Prof. Indranil Sengupta, IIT Kharagpur

Series: notes-2 (transcript-grounded) — Introduction to ethical hacking, OSI/TCP-IP, IP addressing, TCP/UDP, subnetting.

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L01: Introduction to Ethical Hacking

Transcript: [01-introduction-to-ethical-hacking.md](#)

Video: <https://www.youtube.com/watch?v=eoo7MI0lyBc>

Week / theme: Week 1 — Intro, networking fundamentals, TCP/IP

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- Now, in this lecture as I told you, firstly we shall try to tell you what is ethical hacking?.
- There is a related terminological penetration testing, we will also be discussing about that.
- And some of the roles of an ethical hacker, what an ethical hacker is expected to do and what he or she is not expected to do that we shall try to distinguish and discuss.

- Well, ethical hacking as per the definition if you just look at it, it essentially refers to locating the weaknesses and vulnerabilities.
- So, this ethical hacking is the act of locating weaknesses and vulnerabilities in computers and information system in general, it covers everything, it covers networks, it cover databases, everything.

What this lecture teaches

Core concepts from the lecture

I would like to welcome you to this course on Ethical Hacking.

This is the first lecture of this course.

Now, in this lecture, I will try to give you a very overall idea about what ethical hacking exactly is, what are the scopes of an ethical hacker and towards the end, I shall give you some idea about the coverage of this course — what are the things we are expected to cover ok.

So, the title of this lecture is Introduction to Ethical Hacking.

Now, in this lecture as I told you, firstly we shall try to tell you what is ethical hacking?

There is a related terminological penetration testing, we will also be discussing about that.

And some of the roles of an ethical hacker, what an ethical hacker is expected to do and what he or she is not expected to do that we shall try to distinguish and discuss.

So, let us first start with the definition of ethical hacking.

Continued

What exactly is ethical hacking?

Well, we all have heard the term hacking and hacker essentially the term has been associated with something which is bad and malicious.

Well, when we hear about somebody as a hacker, we are a little afraid and cautious ok.

I mean as if the person is always trying to do some harm to somebody else to some other networks, try to steal something, trying to steal something from some IT infrastructure and so on and so forth.

But ethical hacking is something different.

Well, ethical hacking as per the definition if you just look at it, it essentially refers to locating the weaknesses and vulnerabilities.

It means suppose you have a network, you have an organizational network, you have an IT, IT infrastructure, you have computers which contains some software, some data, lot of things are there.

Now, you try a, I mean here you are trying to find out, whether your infrastructural network does have some weak points or vulnerabilities through which an actual hacker can break into your system, into your network.

Continued

So, this ethical hacking is the act of locating weaknesses and vulnerabilities in computers and information system in general, it covers everything, it covers networks, it cover databases, everything.

But how this is done, this is done by mimicking the behaviour of a real hacker as if you are a hacker, you are trying to break into your own network, there you will get lot of information about what are the weak points in your own network.

So, this term is important, by replicating the intent and actions of malicious hackers, whatever malicious hackers do in reality, you try to mimic that, you try to replicate that ok.

Your objective is to try and find out the vulnerabilities and weak points in your network.

Well, you have a good intent, you try to identify the weaknesses and later on maybe the organization will be trying to plug out or stop those weaknesses, so that such attacks cannot occur or happen in the future ok.

This ethical hacking is sometimes also referred to by some other names, penetration testing is a well-known terminology which is used — a phrase, intrusion testing, red teaming, these are also terminologies which are used to mean the same thing.

Well, you can understand penetration testing, the literal meaning of this phrase is, you are trying to penetrate into a system; you are trying to penetrate into a network, you are testing and find out whether or not you are able to penetrate.

And if you are able to penetrate which are the points through which it is easier to penetrate, these are the objectives ok, all right.

Continued

So, talking about ethical hacking, there are some terminology, let us see.

Well ethical hackers are the persons who are actually carrying out ethical hacking.

Now, they are not some unknown entities, they are some organization or persons who are actually hired by the company.

The company is paying them some money to do a penetration testing on their own network and provide them with a list of vulnerabilities, so that they can take some action later on ok.

So, these ethical hackers are employed by companies who typically carry out penetration testing or ethical hacking.

Penetration testing, as I had said is an attempt to break into a network or a system or an infrastructure.

But the difference from malicious attempt is that this is a legal attempt.

The company has permitted you to run the penetration testing on their own network for the purpose of finding the vulnerabilities.

Continued

So, this is a legal attempt, you are trying to break in and you are trying to find out the weak links.

Well, in penetration testing per se what the tester will do, tester will basically generate a report.

The report will contain a detailed report; it will contain all the known vulnerabilities that have been detected in the network as a result of running the penetration testing process ok.

But normally they do not provide solutions.

Well, you can also seek solutions for them, but everything comes with an extra or additional charge right.

So, in contrast, security test is another terminology which is used, which includes penetration test plus this kind of suggestions to plug out the loopholes.

So, this includes in addition analyzing the company security policies and offering solutions, because ultimately the company will try to secure or protect their network.

Of course, there are issues, there may be some limited budget.

Continued

So, within that budget whatever best is possible that have to be taken care of or incorporated.

So, these are some decisions the company administration will have to take fine.

So, some of the terminologies that we normally use hacking, hacking broadly speaking, we use this term to refer to a process which involves some expertise.

We expect the hackers to be expert in what they are doing.

At times we also assume that hackers are more intelligent in the persons, than the persons who are trying to protect the network.

This assumption is always safe to make that will make your network security better ok.

Cracking means breaching the security of a some kind of system, it can be software, it can be hardware, computers, networks whatever, this is called cracking, you are trying to crack a system.

Spoofing is a kind of attack, where the person who is, who is attacking is trying to falsify his or her identity.

Continued

Suppose, I am trying to enter the system, but I am not telling who I am, I am telling I am Mr.

X is somebody else right.

So, it is the process of faking the originating address in a packet, a packet that flows in a network is sometimes called a datagram ok.

So, the address will not be my address, I will be changing the address to somebody else's address, so that the person who will be detecting that will believe that someone else is trying to do whatever is being done ok.

denial-of-service is another very important kind of an attack which often plagues or affects systems or infrastructures.

Well, here the idea is that one or a collection of computers or routers or whatever you can say, a collection of nodes in the network, they can flood a particular computer or host with enormous amount of network traffic.

The idea is very simple, suppose I want to bring a particular server down, I will try to flood it with millions and millions of packets, junk packets, so that the server will spend all of its time filtering out those junk packets.

So, whenever some legitimate requests are coming, valid packets are coming, they will find that the service time is exceedingly slow, exceedingly long, this is something which is called denial-of-service.

Continued

And port scanning is a terminology which you use very frequently, well ports in a computer system this we shall be discussing later.

Ports indicate some entry points in the system which connects the incoming connections to some programs or processes running in the system.

Say means in a computer system there can be multiple programs that are running, and these programs can be associated with something called a port number ok.

Whenever you are trying to attack a system, normally the first step is to scan through some dummy packets ping, these are called ping packets and try to find out which of the port numbers in the system are active.

Suppose, you find out that there are four ports which are active then normally there is a well documented hacking guideline which tells you that for these four ports what are the known vulnerabilities and what are the best ways to attack or get entering those into the system through these ports.

So, this port scanning is the process of identifying which are the active ports which are there and then searching for the corresponding vulnerabilities, so that you can exploit them ok.

These are called exploits, once you identify the ports you try to find out an exploit through which you can get entry into the system, this is roughly the idea.

Now, talking about gaining access into the system, there are different ways in which you can gain access to a system.

Continued

One is you are entering the system through the front door.

So, the name is also given front door access.

Normally, a system, normally I am talking about whenever you try to access the system you try to log in, you are validated with respect to some password or something similar to that.

So, passwords are the most common ways of gaining entry or access to a system in the present day scenario ok.

So, the first attempt through that front door channel will be to guess valid password or try and steal some password.

There are many methods that are used for this purpose.

During this course you will be seeing some of the tools through which you can try and do this ok.

This is the front door.

Continued

The second thing is a back door which normally a person coming is not able to see, but it is there.

Those of you who know there is a back door, they can only enter through that back door.

This is the basic idea.

So, back doors are some you can say entry points to a system which had deliberately kept by the developers.

Well, I am giving an example suppose I buy a router, a network router from some company, they give me some root password and access rights, I change the root password.

So, I am quite happy that means, I have sole access to it, I have changed the password, I am safe.

But sometimes it may happen if something goes down, the company might automatically modify or configure, reconfigure the router through that back door.

They will not even ask you at times.

Continued

They will automatically enter the router through that backdoor entry, there will be some special password through which they can possibly enter and they can make some changes inside.

Such back doors are known to exist in many systems, not only hardware systems also many of these software systems, software packages ok.

Well, usually developers keep it as debugging or diagnostic tools, but sometimes these are also used for malicious purposes ok.

Then comes the Trojan horses.

Now, if you remember the story of the Trojan horse where it is something which was hidden inside a horse, some warriors were hidden inside a horse.

Suddenly some time one night, they just comes out and start creating havoc.

Trojan horse is also in terms of a computer system something very similar.

Here let us think of a software first.

Continued

So, it is a software code that is hidden inside a larger software.

Well, as a user you are not even aware that such a Trojan is there inside the software ok.

Now, what happens sometimes that Trojan software can start running and can do lot of malicious things in your system.

For example, they can install some back doors through which other persons or other packets can gain entry into your system.

Nowadays, you will also learn as part of the course later, Trojans can also exists in hardware.

Whenever you built a chip, you fabricate a chip, without your knowledge, some additional circuitry can get fabricated which can allow unauthorized access or use of your chip, of your system during its actual runtime ok.

And lastly come software vulnerabilities exploitation.

Well, when a software is developed by a company, that software is sold, with time some vulnerabilities might get detected.

Tools mentioned

- **dig**

Commands / syntax from the lecture

- dig into the environment and find out whatever relevant information is required

Key terms

Term	Meaning in this lecture
Cracking	breaching the security of a some kind of system, it can be software, it can be hardware, computers, networks whatever, this is called cracking, you are trying to crack a system.

Exam-oriented recap

- I mean as if the person is always trying to do some harm to somebody else to some other networks, try to steal something, trying to steal something from some IT infrastructure and so on and so forth.
- So, this term is important, by replicating the intent and actions of malicious hackers, whatever malicious hackers do in reality, you try to mimic that, you try to replicate that ok.
- Well, you have a good intent, you try to identify the weaknesses and later on maybe the organization will be trying to plug out or stop those weaknesses, so that such attacks cannot occur or happen in the future ok.
- So, in contrast, security test is another terminology which is used, which includes penetration test plus this kind of suggestions to plug out the loopholes.
- So, this includes in addition analyzing the company security policies and offering solutions, because ultimately the company will try to secure or protect their network.
- This assumption is always safe to make that will make your network security better ok.
- Cracking means breaching the security of a some kind of system, it can be software, it can be hardware, computers, networks whatever, this is called cracking, you are trying to crack a system.
- Spoofing is a kind of attack, where the person who is, who is attacking is trying to falsify his or her identity.

L02: Basic Concepts of Networking — Part I

Transcript: [02-basic-concepts-of-networking-part-i.md](#)

Video: <https://www.youtube.com/watch?v=-iShap4w4S8>

Week / theme: Week 1 — Intro, networking fundamentals, TCP/IP

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- Hello welcome back, as we have discussed in the first lecture, in this course on Ethical Hacking, we shall be talking about a lot of issues and techniques and tools related to computer networks.
- So, it is imperative that some basic background of computer networks and some of the relevant aspects of it should be discussed as part of the course.
- So, in this lecture specifically we shall be starting with a discussion on some Basic Concepts of Networking.

What this lecture teaches

Core concepts from the lecture

Hello welcome back, as we have discussed in the first lecture, in this course on Ethical Hacking, we shall be talking about a lot of issues and techniques and tools related to computer networks.

So, it is imperative that some basic background of computer networks and some of the relevant aspects of it should be discussed as part of the course.

So, in this lecture specifically we shall be starting with a discussion on some Basic Concepts of Networking.

In this lecture, we shall be very specifically looking at different types of computer networks; the difference between the circuit switching and packet switching mode of data transfer and one method of packet switching data transfer which is called virtual circuits.

So, we shall be talking about these few things.

So, let us start with some of the basic concepts on computer networking ok.

Now, as you already know that a computer network is essentially a communication system.

Why do you use computer networks?

Continued

We use computer networks, so that different computing devices, nowadays the computing devices appear in various shapes and forms, not necessarily they are computers.

They can be anywhere, of any type that can be some gadgets, they can be some equipments anything.

The idea is that they are connected to the internet or some kind of a network and they can communicate with other devices.

You think of a security system in your home, that is also connected to some kind of a network, connected to some central servers, so that whenever some, you can say anomalous incident occurs, some central server will be notified accordingly ok.

So, essentially a computer network we can define as a communication system, where some kind of computers or there any kind of hosts, as I told you, the nodes that are connecting are not necessarily computers.

They are some devices, which have communication capability; they can be connected to the network right.

Now, the question is why do we need networking?

The reason is very clear; we want to have better connectivity.

Continued

So, that the nodes remain connected with each other, we can have better communication.

And obviously, sharing of resources nowadays, we talk about cloud computing, where not only storage space, but also computing resources are available online on the network, on the cloud.

So, we can share some valuable resources over the network and it helps in bringing people together as we are seeing with our social networking applications that are coming up.

Talking about the different types of computer networks, very broadly speaking, we can classify computer networks as Local Area Networks or LAN and Wide Area Networks or WAN.

Of course, there are some intermediate nomenclatures that are also used like, people talk about metropolitan area network, which is somewhere in between, but we are not going into that detail.

Now, essentially local area network as the name implies, they are used to connect hosts, which are separated by relatively small geographical area.

Now, this relatively small is a, really a relative term, it can be in the same room, it can be in the same building or maybe within the same campus.

Means, whether or not it is a local area network, it depends upon the technology that is being used.

Continued

Nowadays, the most commonly used local area network standard is the Ethernet, various versions of Ethernet are there.

It started with a very primitive form, the 10 mega Hertz, mbps, megabits per second Ethernet.

Subsequently nowadays we have several giga Hertz speed, 100 giga Hertz and also beyond such speeds are coming up ok.

Now, local area networks broadly speaking, this is the fastest kind of network and it is also relatively cheaper in the long run.

Fastest when I talk about fast nowadays, by default we have something called gigabit per second speed available on our desktops, but there are technologies with which you can even enhance the speeds 10 gbps, 40 gbps, 100 gbps those technologies are also available in LAN.

Talking about wide area network, they are used to connect devices or networks which are widely separated.

Now by meaning widely separated, they can be across campuses, they can be across cities, countries or even continents.

Traditionally such wide area networks were relatively slow.

Continued

So, that is why we are saying that they are slower as compared to local area network.

And they are also expensive in the long run, well initial deployment investment may not be that high, but you will have to pay a very hefty rental, annual rental or monthly rental to use this wide area network facility.

Because, there is some kind of a service provider you are subscribing to the service, you will have to pay for the service, but in contrast a local area network is something which is your own, you are buying the equipments, you are buying the cables and the entire management the entire infrastructure is your own.

So, you are not paying any body for using that infrastructure right.

Now, talking about data communication over a network, let us look at rough picture like this, let us say these ABCD are the different nodes in a computer network and let us suppose this node A wants to send some message or some data to this node H and this BCD etc., these are so called intermediate nodes.

So, from A to H there is no direct path, whatever data you want to send, they have to traverse or travel through various other intermediate nodes.

Now in computer network terminology these intermediate nodes are typically routers, we shall be looking into routers again later.

Broadly speaking communication may happen using two techniques; one is called circuit switching which was more traditional the old method and nowadays we have something called packet switching, let us look into this one by one.

Continued

Talking about circuit switching, now whenever you want to communicate between two nodes, again I am talking about this A and H, what we are saying is that we are kind of establishing a circuit or a dedicated path for the communication.

If you think of our old analogue telephone systems, when we used to dial a number, then a dedicated copper path used to get established from the caller to the person you are calling.

Similar is the case here, say whenever A wants to communicate with H, some dedicated path, here it is marked in red, you can see via C, D and G possibly, this path will get established.

And this path will be dedicated means, during the time of data communication, this dedicated path will be used for sending this information chunks, bits or whatever you are sending.

And as you see, the path that we are selecting or we are deciding upon, they will follow a fix sequence of intermediate nodes or links through C, D and G ok.

Now, on every physical link, this is a link, this is a link, this is a link and this is a link, there are 4 communication links, that exists in between right.

So, over each of these communication links, something called a logical channel gets defined or established.

Logical channel is something which guarantees you some dedicated bandwidth.

Continued

Suppose you want a dedicated bandwidth of 64 kilobits per second.

Let us take an example.

So, all these communication channels that you are establishing, they will be guaranteeing a minimum bandwidth of 64 kbps ok, that is how this works?

That is not that you are actually physically using that link means, on your own you are not allowing others to share, not like that, maybe this particular link is also shared by some other communication, but for this particular communication for from A to H, there is a 64 kilobyte channel or bandwidth that you are guaranteed to get, you have been provided with right that is the idea ok.

So, in circuit switching, because we have a path to be established between the sender and receiver before you can actually start the communication, there are broadly three steps that need to be followed.

The first is clearly a step of connection establishment, just like when you are dialing a telephone that step is the connection establishment phase, where the dedicated channels or links are getting established.

This is required before the data transmission can start and once you have connection established, you can actually carry out the data transfer.

Now, the advantage of circuit switching comes in the second step, when data transfer takes place, it is in the maximum possible speed without any disruption.

Continued

So, you have dedicated bandwidth along all the links.

So, that amount of dedicated bandwidth is always available to you ok.

And of course, at the end when you are finished with the data communication, you terminate the connection.

Terminate the connection means the resources that you had reserved, the communication links or the bandwidths, now, you can release them, now some other communication can use that bandwidth now ok.

This is what do you mean by resources we allocating, we are deallocating ok, fine.

Now, circuit switching has some obvious drawbacks.

First drawback is that because you are having dedicated links from the source to the destination, during the time, the communication is active, this entire channel is dedicated for the communication.

But you see, you, when you send a, transmit data it is not that data are being transmitted continuously.

Continued

Usually computer data are bursty in nature, sometimes there is lot of data and there is a period where there is no data, sometimes again there is lot of data flowing, sometimes again there is no data.

So, if you would reserve the channel for the entire period, your average channel utilization may not be that good, this is the drawback ok.

Because, you are dedicating or reserving the channel capacity for the entire duration of communication, this will be quite inefficient for data traffic, which is typically generated by computer, which is bursty in nature.

Data appears in burts, in between there is no data, there is a gap.

But for voice communication when we talk over telephone, we normally do not remain silent, either I am talking or the person on the other side is talking, there is always some communication going on with respect to voice traffic.

Nowadays most of the voice communication is also being carried out through computer network kind of an infrastructure.

They are converted to digital, digitized and then voice is transmitted as data ok.

And the other drawback is that, because of the connection establishment requirement, there will be an initial delay, only after that the communication can start ok.

Continued

These are some of the drawbacks you have to remember.

Next let us come to the other method so called packet switching.

Now, packet switching as I said, this is more widely used nowadays, this is the modern form and this is used for long distance data communication, because it has some advantages.

The main advantage is that you are not dedicating any network resources here.

All communication links are shared, the links that your data will be traversing, they are all shared; they are not dedicated only for your communication ok.

Because, the links are shared, there is no concept of guaranteed bandwidth typically.

Typically, there is no such concept, but of course, you can have something called quality of service, you can have a guarantee on that, but typically there is no guarantee on the bandwidth.

So, when there is lot of traffic on the network, you may feel that your data transmission or communication has become slow.

Continued

So, this basic concept of packet switching as evolved over time, we shall be looking at this in some more details.

Now, in packet switching, the essential idea is that when you are trying to send a message, suppose, here we have a message that you want to send.

First step, the message is broken up into smaller pieces or chunks which are called packets; these are my packets 1, 2 and 3, there are 3 packets ok.

And with each packet you add a header.

This header will contain some relevant information like, where you want to send the packet, destination address and some information relating to that.

So, these packets, you are dividing your message into each of these packets will be having some additional overheads in terms of headers ok.

And each of these packets are transmitted separately.

The message is not transmitted as a whole rather the packets that constitute the message, they will be transmitted one by one, this is the basic idea behind packet switching.

Continued

So, it is not the message, you are thinking about is being transmitted, but rather the individual packets that are getting transmitted all right.

There is another concept here that is used, this is a very important concept, which is called store and forward concept.

The idea is as follows, let us say, I have a node here, I have node here, I have a node here.

So, I am sending some data first to here and this fellow is forwarding the data to this fellow.

Now, if we think of this central node.

What happens is that, whatever packets that are being transmitted first through this first link, they will get stored in an internal buffer.

There will be a buffer, a memory space where these packets will get stored right.

And later on from this buffer, these packets will get forwarded to the next link.

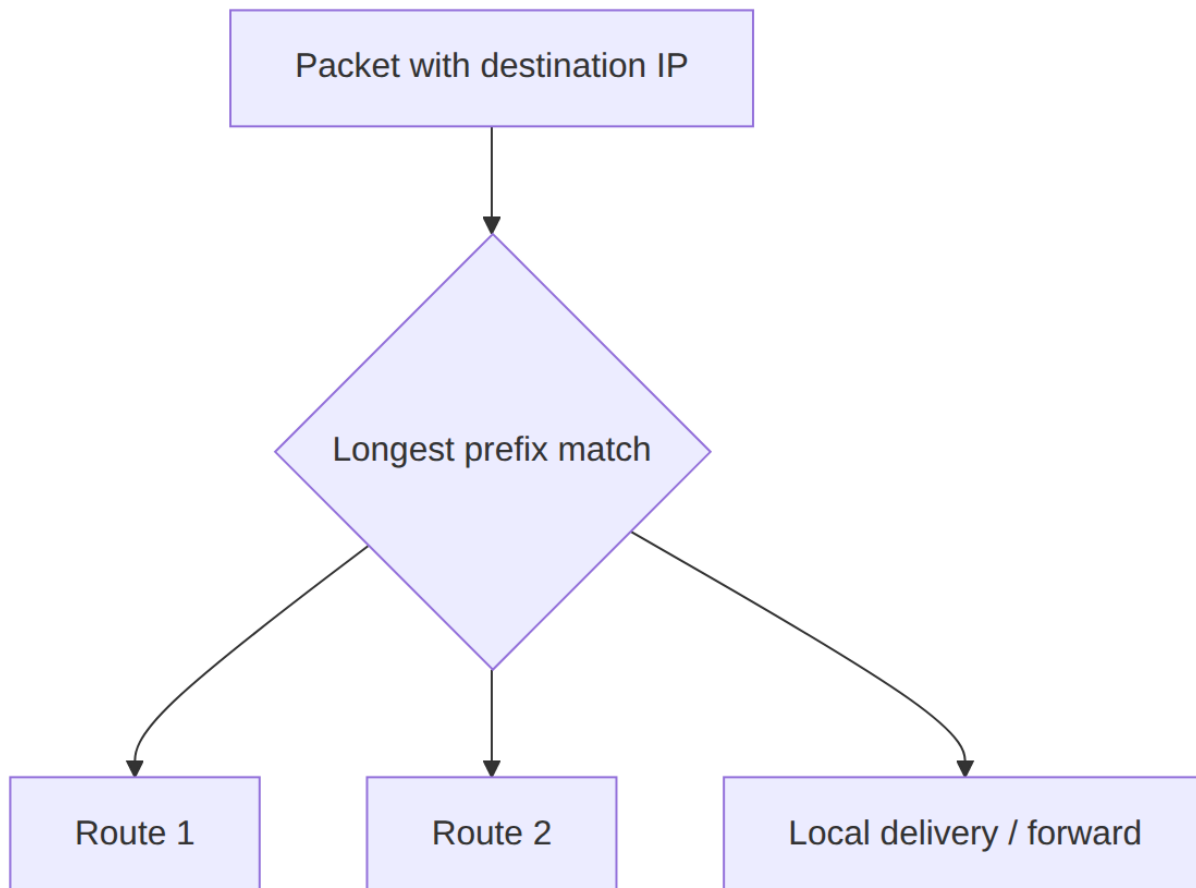


Figure 1: Architecture diagram

Key terms

Term	Meaning in this lecture
Terminate the connection	the resources that you had reserved, the communication links or the bandwidths, now, you can release them, now some other communication can use that bandwidth now ok.
Now, this information is typically kept in a place which	a routing table.

Exam-oriented recap

- Talking about circuit switching, now whenever you want to communicate between two nodes, again I am talking about this A and H, what we are saying is that we are kind of establishing a circuit or a dedicated path for the communication.
 - Similar is the case here, say whenever A wants to communicate with H, some dedicated path, here it is marked in red, you can see via C, D and G possibly, this path will get established.
 - Now, the advantage of circuit switching comes in the second step, when data transfer takes place, it is in the maximum possible speed without any disruption.
 - So, that amount of dedicated bandwidth is always available to you ok.
 - But for voice communication when we talk over telephone, we normally do not remain silent, either I am talking or the person on the other side is talking, there is always some communication going on with respect to voice traffic.
 - These are some of the drawbacks you have to remember.
 - Now, packet switching as I said, this is more widely used nowadays, this is the modern form and this is used for long distance data communication, because it has some advantages.
 - The main advantage is that you are not dedicating any network resources here.
-

L03: Basic Concepts of Networking — Part II

Transcript: [03-basic-concepts-of-networking-part-ii.md](#)

Video: <https://www.youtube.com/watch?v=jp0JdeeFUG0>

Week / theme: Week 1 — Intro, networking fundamentals, TCP/IP

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- So, we continuing with our discussion on the Basics of Computer Networking.
- Let us look at the datagram approach, well, if you compare with the virtual circuit approach, the main difference comes here that no route is established beforehand.
- So, this is the second part of this lecture on basic concepts of networking.
- Now, if you recall in our last lecture, we had talked about the broad approaches of data transmission over a network.
- Today we shall be starting with the other alternative which is the so called datagram approach.

What this lecture teaches

Core concepts from the lecture

So, we continuing with our discussion on the Basics of Computer Networking.

So, this is the second part of this lecture on basic concepts of networking.

Now, if you recall in our last lecture, we had talked about the broad approaches of data transmission over a network.

We are talking about something called packet switching, we mentioned the virtual circuit approach, where some dedicated path gets established, a fixed path get established called a virtual circuit, all packets will be following that path.

Today we shall be starting with the other alternative which is the so called datagram approach.

So, in this lecture we shall be talking about first the datagrams, some issues relating to that and then we shall be talking about layered network architecture, which is a very commonly used popular architecture for designing networking systems ok.

Let us look at the datagram approach, well, if you compare with the virtual circuit approach, the main difference comes here that no route is established beforehand.

This is one difference or advantage you can say, you have here advantage in the sense that there is no initial delay.

Continued

In virtual circuit because of the route establishment, there was an initial delay and once the route is established, then only the packets can be transmitted.

Now, here because there is no route established, so each packet is transmitted as independent entities.

Just like you think of a postal system as an analogy, I am talking about a postal system.

When you drop a letter in a letter box, each letter is sent or is being you can say despatched as an independent entity.

So, you do not know which path the letter is following or actually how it is going ok.

So, that you do not know, but one thing you know is that, that when you post a letter, you specify some kind of an address, their destination address.

So, here also when you transmit a packet, the packet must carry the address of the destination node along with it.

Because, it is being sent as an independent entity, you must have the address ok.

Continued

And, it does not maintain any kind of history like, there is no concept of a path that is established and so on.

So, in that sense, there is no history of data communication or transmission that is maintained in this case.

Each packet is being routed as an independent entity fine.

So, here the difference is, here every intermediate node will be taking some dynamic decisions, it will be maintaining something called a routing table.

So, you see if you think of a scenario, where there is a node A, there is a node B, and there is a node C.

So, you are sending data from node A to node B and then node B to node C.

Now, this node B will be having some kind of a table that it maintains, the table contain some relevant information.

And what kind of information it contains?

Continued

It contains information like, like whenever an incoming packet comes, so, it will contain information like, where this packet has to be forwarded to ok.

There can be multiple outgoing links from B and to take this decision, what is required is that when you are thinking about 1 packet, I mentioned that each packet will be containing a header.

In this header there must be a source address and also a destination address.

Just like a letter, when you post a letter, you give the address where you want to send the letter to and in case the letter cannot be delivered, you also give your own address, so that the letter can come back to you.

In a same way in the header there are other information also, but the most important one is source address, the person who is sending and the destination address where you are sending right.

Now, just like the normal postal system, here there can be some problems.

Like, when you post letters, there is no guarantee that the order you are posting the letters, the letters will get delivered in the same order, there is no guarantee regarding that, courtesy of our postal system or the way the letters are sorted and delivered.

So, packets may be delivered out of order, why?

Continued

Because each packet is being routed as an independent entity, there is no guarantee that all packets are flowing or following the same path.

Some packet may be following a shorter path, some packet may be following a longer path.

So, the delays of each packet can be different ok.

So, the delivery of the packet can be out of order.

Similarly, if some of the intermediate node temporally goes down, all packets which are stored in buffer, will get lost.

So, some packets might get lost also.

This also happens in a postal system, sometimes a letter never gets delivered, but something which happens here, which is not there in a postal system, is that, sometimes duplicate packets may get generated.

Like for every link, suppose there is a link between a node to the next node.

Continued

This node will be sending some packets to the next node and will be expecting some acknowledgement to come, that well I have received the packets correctly.

If the acknowledgement does not come, this sender will assume, there is some something wrong, let me send it again ok.

But it may so happen that because of some other problem, the acknowledgement got delayed, but if this sender sends the same packet again, there will be duplicate which will be generated right.

So, some duplicate packets might get generated ok.

Now, talking about the advantages of this approach, datagram approach; first thing is that if the number of packets is not very large, this will be faster than the virtual circuit approach, because in the virtual circuit approach, there is an initial overhead of connection establishment.

But once the connection establishment is over, packet transmission will be very fast.

So, if there are large number of packets, then virtual circuit can be faster overall ok.

Datagram approach does not need any route establishment in the beginning or termination at the end, it does not require these.

Continued

So, in that sense it is more flexible.

Packets can follow different paths, if sum of the links get overloaded, then the packets can follow an alternate path, these flexibilities are there ok.

So, this as I mentioned, it can handle congestion in the links or some of the links may fail altogether, then an alternate link may be used which was not there for a virtual circuit all right.

Now, if you just compared these approaches which we have talked about, now for comparison we usually compare with respect to some parameters; one is called propagation delay.

Like you see when we talk about a communication between two nodes; propagation delay is the signal delay from source to the destination.

How much time a single signal transition takes to move from this place to this place.

If, it is a copper link, it can be something, if it is a fibre optic link, it can be different, if it is a satellite link, it can be different.

So, depending on the type of link, how much time suppose you send one bit of data?

Continued

How much time that one bit will take to reach the other end, it depends on the medium.

Suppose, you are using a satellite, then it will be taking a fraction of a second to go up to the satellite and again come back ok.

So, it depends on the medium, but transmission time is different.

Transmission time depends on the bandwidth of the link, well you know that links are categorised by the speeds, kilobytes per second, megabits per second, gigabits per second.

Suppose it is a 1 megabit per second, link 1 Mbps.

So, 1 million bits can be sent per second.

So, if there are 1000 bits you are trying to send, so, here how much time will it take?

Here every bit will take 1 microsecond 1 by mega.

Continued

So, 1000 bits will take one millisecond that is the transmission time.

It depends on the size of the message, how many bits are there and the bandwidth of the link.

It does not depend on the propagation delay; whether it is a satellite link or a copper link whatever.

So, these two are two separate things and finally, processing delay.

If your communication is going via intermediate nodes and you know, here we use something called store and forward approach.

So, these intermediate nodes will also take some time to receive a packet, store it, consult the routing table, take some decision and then forward it.

So, it will take some small time for that.

So, that is the processing delay.

Continued

So, these are the different components of delay in a typical network communication.

Now, talking about circuit switching, here only this initial circuit establishment phase is there, but once this phase is over, this delay is over; you have a dedicated connection from source to destination, data transmission can proceed in the maximum speed continuously without delay.

Because, you know, you have a guaranteed bandwidth, you can send the bits continuously.

Now, talking about virtual circuit, well during the initial virtual circuit establishment phase a special packet is usually sent, I am not going into the detail, which is called call request packet.

This call request packet is responsible for establishing the virtual circuit and setting up the routing tables of all the intermediate nodes, corresponding to this particular virtual circuit number ok.

Now, once the virtual circuit has been established and acknowledgement signal or packet will be traversing along the same path back, this is called a call accept packet.

So, that the sender and all the intermediate nodes will know that the circuit has been established.

Now, the packets can flow, now the packets can be sequentially one after the another in a pipeline fashion using store and forward approach, this is how virtual circuit will work.

Continued

Now, talking about the datagram packet switching, because there is no connection establishment phase, so there is no initial delay ok.

The packets are sent out independently, may follow different path and this also follows store and forward concept, but if the number of packet is very large, then the total time required here maybe more as compared to virtual circuit, but here you have lot of flexibility.

You can share the links, if there are some link failures, link congestion, you can overcome those problems.

So, those flexibilities are there and because of that, this datagram approach is the most widely used approach in the internet today ok.

Now, let us talk about something called layered network architecture.

Now, see a networking system has a number of responsibilities.

Broadly speaking it is data communication, but if you look into the detail, there are a number of steps which are involved in the process.

Now, I mean in order to design a networking system in a flexible way, a conceptual layering has come in.

Continued

Layering means you divide the total functionality into a number of levels, where the function of each of the level is very well defined ok.

Now, a very well-known model is the so, called OSI model.

This is open systems interconnection model, this was proposed by international systems organisation, ISO.

This is called ISOs OSI model, it is called, there are 7 layers in this model.

I shall show the layers very shortly and there is a hierarchical relationship among these layers.

The main objective of this layering is, as I mentioned, if you divide the functionality into well-defined layers, it becomes much more systematic.

Your design of the networking software becomes much more systematic.

Not only that you may choose to modify the design of one of the layers without touching the other layers.

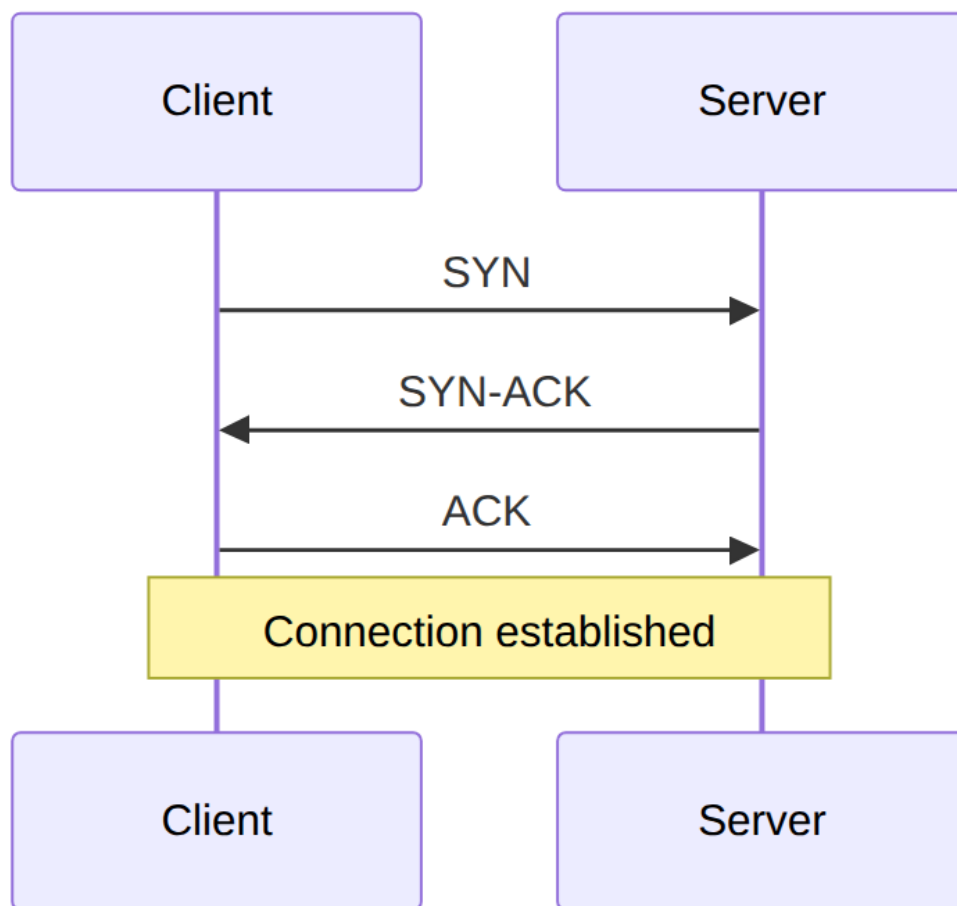


Figure 2: Architecture diagram

Key terms

Term	Meaning in this lecture
Layering	you divide the total functionality into a number of levels, where the function of each of the level is very well defined ok.
This	ISOs OSI model, it is called, there are 7 layers in this model.
Point to point	2 nodes which are directly connected by a link is called a point to point connection.

Exam-oriented recap

- This is one difference or advantage you can say, you have here advantage in the sense that there is no initial delay.
- So, here also when you transmit a packet, the packet must carry the address of the destination node along with it.
- Because, it is being sent as an independent entity, you must have the address ok.
- It contains information like, like whenever an incoming packet comes, so, it will contain information like, where this packet has to be forwarded to ok.
- In this header there must be a source address and also a destination address.
- In a same way in the header there are other information also, but the most important one is source address, the person who is sending and the destination address where you are sending right.
- This also happens in a postal system, sometimes a letter never gets delivered, but something which happens here, which is not there in a postal system, is that, sometimes duplicate packets may get generated.
- These are the some of the advantages.

L04: TCP/IP Protocol Stack — Part I

Transcript: [04-tcp-ip-protocol-stack-part-i.md](#)

Video: <https://www.youtube.com/watch?v=EsJVCgefqnY>

Week / theme: Week 1 — Intro, networking fundamentals, TCP/IP

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- In this lecture, we shall be starting our discussion on something called TCP/IP Protocol Stack.
- Now, in this lecture, I shall be talking about firstly, the TCP/IP protocol stack on which the internet is basically based on.
- So, if a node, you connect to the computer, if it understands this TCP/IP protocol, then it can communicate with the Internet to any other computer in the world right ok.

What this lecture teaches

Core concepts from the lecture

In this lecture, we shall be starting our discussion on something called TCP/IP Protocol Stack.

So, the title of the lecture is TCP/IP protocol stack, the first part.

Now, in this lecture, I shall be talking about firstly, the TCP/IP protocol stack on which the internet is basically based on.

We shall be looking at the basic functions of these TCP, UDP and IP, the most important three components of this TCP/IP stack.

And with the help of an example we shall see the concept of something called data encapsulation.

So, let us start with TCP/IP.

As I mentioned TCP/IP is the most fundamental protocol in the internet.

Just to imagine, what is an internet; internet is a network of networks.

Continued

There are so many networks, they are all connected together, inside a network there are so many different protocols that are being followed, computers are connected to it, but somehow they are all able to communicate with each other.

So, there must be some common language, all these computers are speaking and it is this TCP/IP that is the common language.

So, if a node, you connect to the computer, if it understands this TCP/IP protocol, then it can communicate with the Internet to any other computer in the world right ok.

Now, TCP/IP by virtue of its standardization, this allows computers to communicate and share resources over the internet all across the world.

This is used as an universal standard in the Internet.

Now, this TCP/IP as a universal standard is used to bridge the gap between various different platforms.

Well, when you say platforms, you can think about operating system to start with, you have Windows, you have Linux, you have Mac, then you can have different kind of networks also.

You have Ethernet, you can have ATM, you can have so many other kinds of networks.

Continued

Talking about wide area networks, there also so many standards are there.

So, there are various alternatives and options available at every step.

So, how do you make them work together?

It is this TCP/IP; TCP/IP is the common glove that binds everything together right.

Now, this TCP/IP, the work, the standardization started in the early 1970s as part of a project which was initially funded by the US military.

But subsequently, it spread to the different universities and finally, it became a de facto standard in the Internet.

Nowadays everybody uses TCP/IP.

Now, in TCP/IP, the network layering, well you have already seen in the previous lecture the 7-layer OSI model which was proposed by International Standards Organization.

Continued

Therefore, TCP/IP, there is a similar kind of layering which is followed, but it is somewhat simplified, it does not contain all the 7-layers.

In fact, it uses a very much simplified 4-layer model, there are only 4-layers here ok.

So, in contrast to the 7-layer OSI model, TCP/IP has only 4-layers.

Let us try to see this layering side by side and try to make a comparison ok.

Here you see the 7-layer model on the left, and the 4-layer TCP/IP model on the right.

And the 7-layer model as I mentioned, the lower 3-layers are the point-to-point layers and the upper 4-layers are host-to-host layers.

Then the TCP/IP, what they have done, they have simplified that the all upward four host-to-host layers, they have integrated in a single application layer ok.

You see it depends on the application whether you actually require a session, whether we actually need to do some changes in your data presentation.

Continued

So, it really depends.

So, instead of keeping separate layers, the TCP/IP has integrated all of them in a single layer and calling it the application layer.

And this transport layer this of course, has a more or less correspondence with the transport layer here.

This is a host-to-host layer, this is responsible for end-to-end message transfer.

Network layer also has a correspondence.

Network layer is mainly responsible for packet routing, delivery of packet across the Internet, across the network.

And again this data link and the physical layers, they are merged together and they are referred to as a single link, sometimes this is referred to as a data link or a physical link, they are called differently.

The point is that, in a computer system when you, when you want to connect it to a network, there is some kind of a network card, you want to use network interface card, and the network interface card typically integrates both data link and the physical layers together that is why in a simplified TCP/IP model, the two layers have been merged into a single layer ok.

Continued

So, this is the simplified 4-layer version in TCP/IP.

Now, in this 4-layer model, when you think of the data transmission, again it is somewhat simplified as you can see here.

This A is the sender, C is the receiver, and B is an intermediate node.

So, this A generates some data from some application; from application it goes down to the transport layer; from transport layer it goes down to the network layer, network layer to the data link layer and to the data link layer it sends the data actually over the data link or the point to point link.

Now, this intermediate node B receives the frame here, sends it up to the network layer; network layer takes a decision where to forward the packet next, and accordingly it sends it down to the data link layer of the outgoing link.

This B may be having several outgoing links, there will be one data link corresponding to every outgoing link.

So, it will be sending out to the data link corresponding to the correct outgoing link and the frame will be going there, received, and it will be traverse in the reverse direction, and finally, took some application in the destination C.

Now, you can see, this transport is a, this is an end to end layer or host-to-host layer as if A and C are communicating directly.

Continued

This intermediate node B is not visualized at this layer, and also at the application layer ok.

These are called end to end or host-to-host layers.

Now, talking about the TCP/IP protocol suite; TCP/IP protocol suite is not a single protocol, but rather it is a family of protocols or collection of protocols that is why it is called a family of protocols.

And these protocols are all based on datagram mode of data communication, they do not rely on virtual circuits, they rely on datagrams.

So, with and all these things we have already discussed earlier.

Datagrams are sent independently from one node to another as a sequence of datagrams, they are all independent ok.

And the same message can be broken up into several packets, and each of this package or datagrams may follow different routes, so may follow different routes.

So, delay may be variable, the arrival order at destination may be different, this already we have discussed earlier all right.

Continued

Now, this is a diagram which shows some of the TCP/IP family member, there are more in fact, so the most important ones I showed here shown here.

So, in the lowest level this is the data link layer, sometimes it is called the data link and hardware layer as I said physical layer different names.

So, if you have an Ethernet connection, you will be using an Ethernet layer here, this will be at Ethernet network card that will constitute the lowest layer.

This is your network layer, this is here you have the network layer.

Now, at the network layer, the most important protocol is the internet protocol or IP.

IP is the most important protocol that is responsible for routing of the packets right.

Now, in addition, there are some other protocols, I will briefly talk about this, after this ICMP, IGMP, ARP, RARP, I will talk about these.

They all work at the network layer level.

Continued

At the higher layer transport layer there are two alternate protocols TCP and UDP, Transmission Control Protocol and User Datagram Protocol.

And at the topmost layer, application layer, you can have any arbitrary user applications running.

Now, in addition as part of the TCP/IP, there are some specific applications also which are also included in the family like file transfer protocol, trivial file transfer protocol, simple mail transfer protocol, all our mail email is send and receive using this SMTP protocol, Simple Network Management protocol, domain name system and so on, there are many others ok.

So, this is just a very quick picture.

Now, talking about these four, I will very briefly talk about these.

This Address Resolution Protocol or ARP, this is basically used to convert the IP address, well each computer is having an IP address, will have an IP address.

This IP address is converted to an hardware address which is called a MAC address; MAC address is referred to as a hardware address.

The Reverse Address Resolution Protocol or RARP does the reverse, it converts the MAC address or hardware address to IP address.

Continued

Now, whenever data are flowing in a network in a LAN this ARP and RARP protocols are very useful, they are used quite widely, we will be, I mean will be discussing this later again.

Now, then we have this ICMP or Internet Control Message Protocol which is mostly used for a device on the network to send error message to other members in the network ok, like some service not available, something not found, etc.

And finally, Internet Group Management Protocol, IGMP.

So, here a node can communicate with its adjacent routers, some information about its multicast group membership.

Let whenever it wants to do some broadcast or multicast selectively, it wants to send data to some set of nodes.

So, it can inform the routers that this is the kind of multicast groups that you want to send data or receive data from.

So, these protocols are roughly used for these purposes.

Now, a typically scenario I am showing here.

Continued

Well, here I am showing two paths; one is a TCP path and other I am showing the UDP path.

Well, here the application will be some user process, which will be using TCP at the transport layer level.

TCP will be using IP at the network layer level and IP will be using the lowest layer hardware layer level or you can use UDP, the user process may be communicating with UDP, IP then hardware layer level.

So, what I mean to say is that when you are developing some application, you may either chose to, choose to use TCP or you may choose to use this UDP both options are available to you.

TCP has some features, UDP has some features, depending on your suitability, what exactly you need, you can use either TCP or UDP.

But the point to note is that below TCP and UDP, you have a common member IP.

So, IP is the network layer, it is a basic datagram routing service that is common, but above IP, you have a choice, you can have either TCP or UDP.

Now, very briefly let us here, let us look at the functionality.

Continued

Well, what does IP do, while IP is responsible for routing of the packets or the datagrams, it transports datagrams, sorry datagrams or the packets from a source to a destination.

This is called routing.

So, it will decide which node to forward it the packet next, that next node will again decide where to forward that packet next, this way the packet will finally reach the destination.

So, the network layer or the IP layer in each of these nodes will be responsible for this routing or packet forwarding.

This IP has an additional responsibility will, if it finds that a packet is large, too large, it may break a packet into smaller packets, this is called fragmentation, this we shall see.

The point to note is that IP uses datagrams and we mentioned earlier, when you talked about datagrams is that, datagram is an unreliable service; unreliable in the sense that some datagrams might get lost, some duplicates might get generated, datagrams maybe received out of order at the destination.

So, such scenarios may happen.

So, means if you are using IP, you must be, you must be aware of these problems that some packet may actually get lost fine.

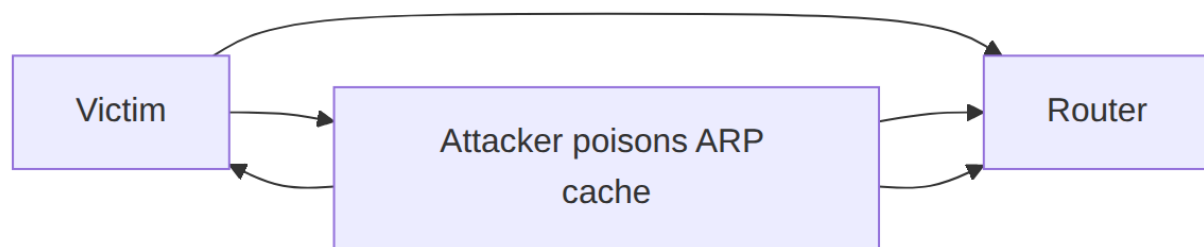


Figure 3: Architecture diagram

Key terms

Term	Meaning in this lecture
This IP address is converted to an hardware address which	a MAC address; MAC address is referred to as a hardware address.
This	routing.

Exam-oriented recap

- We shall be looking at the basic functions of these TCP, UDP and IP, the most important three components of this TCP/IP stack.
 - So, there must be some common language, all these computers are speaking and it is this TCP/IP that is the common language.
 - Now, this is a diagram which shows some of the TCP/IP family member, there are more in fact, so the most important ones I showed here shown here.
 - Now, at the network layer, the most important protocol is the internet protocol or IP.
 - IP is the most important protocol that is responsible for routing of the packets right.
 - Now, whenever data are flowing in a network in a LAN this ARP and RARP protocols are very useful, they are used quite widely, we will be, I mean will be discussing this later again.
 - Let whenever it wants to do some broadcast or multicast selectively, it wants to send data to some set of nodes.
 - So, means if you are using IP, you must be, you must be aware of these problems that some packet may actually get lost fine.
-

L05: TCP/IP Protocol Stack — Part II

Transcript: [05-tcp-ip-protocol-stack-part-ii.md](#)

Video: <https://www.youtube.com/watch?v=atVnsf6PhO0>

Week / theme: Week 1 — Intro, networking fundamentals, TCP/IP

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- So, we continue with our discussion on the TCP/IP Protocol Stack.
- Now, in this lecture we shall specifically be looking at some details about the IP packets, the IP formats, the IP header fields and so on ok.
- Now, another thing I also mentioned, this we shall be discussing later, that IP is also responsible for breaking up a large packet into smaller packets if required.
- Now, in every intermediate node, there will be something called a routing table, these we shall be discussing later in detail.
- First we will indicate a version which is a 4 bit field, which indicates which version of IP you are using.

What this lecture teaches

Core concepts from the lecture

So, we continue with our discussion on the TCP/IP Protocol Stack.

This is the second part of the lecture on TCP/IP protocol stack.

So, recall in our previous lecture we talked about the overall TCP/IP protocol stack architecture, what are the different family members in the TCP/IP, notably TCP, UDP and IP.

Now, in this lecture we shall specifically be looking at some details about the IP packets, the IP formats, the IP header fields and so on ok.

So, let us see talking about IP datagrams.

Now, the IP layer what we mentioned?

We mentioned that in the TCP/IP protocol stack, the IP layer is nothing but the networking layer in the stack.

Now, in the networking layer, the main responsibility is to route the packets that are flowing through the network, this is the main responsibility of the IP layer.

Continued

Now, another thing I also mentioned, this we shall be discussing later, that IP is also responsible for breaking up a large packet into smaller packets if required.

So, broadly speaking the IP layer, it provides a connectionless and unreliable delivery system for packets.

Essentially, it is a datagram service, it is a layer, which provides a mechanism for transmission and routing of datagrams from one node to another in the network in the internet.

Now, in every intermediate node, there will be something called a routing table, these we shall be discussing later in detail.

With respect to the routing table, this IP packets will be coming, they will be compared against and they will be forwarded to one of the outgoing links in a suitable manner.

Now, in the IP layer one thing we mentioned just like datagram, each packet is independent of one another, there is no relationship between successive packets.

So, in that sense the IP layer does not maintain or need not maintain any kind of history and it must be provided with sufficient information, so that it can forward the packet correctly to the final destination.

For that what is required, you need to provide the destination address.

Continued

That must be there in the header and also it must contain the source address, because sometimes, some acknowledgment or some information needs to come back to the sender, that is why both source and also destination address are included as part of the IP address.

And, just like datagram service the IP layer does not ensure reliability and it also does not guarantee delivery of packets, some packets might get lost.

In addition I told you, duplicates might get generated, packets may get also delivered out of order.

So, all these things can happen.

Now, other thing is, this IP layer provides some kind of encapsulation, which in our previous lecture, we have very briefly looked at by taking an example of that trivial file transfer protocol.

We had seen that when a data chunk is moves from the higher layer to the lower layers in the TCP/IP protocol stack, some headers get progressively added.

Now, if you think of the IP layer, this is my IP layer and, some data packet is coming to the IP layer, some data packet, let us call it D.

So, what IP will do, it will add some header to this data packet; it will add some header to this data packet.

Continued

And, IP layer adds minimum of 20 bytes of header, typically 20 bytes only, but in some times, it can be more than 20 rarely, but typically 20 bytes, 20 bytes of header is added to the data.

And this includes source and destination address and some other information also ok, which helps in routing the packet and some other services this we shall see.

So, just this is what I am talking about some data is coming to the IP layer and the IP layer is adding some 20 bytes of header to it.

And is then forwarding it to the lowermost hardware layer or the data link layer whatever you call ok.

This is an overall picture which shows the format of an IP packet.

So, this is the format of an IP datagram.

Where you can see, the first part of it here?

These are the headers and this part is an optional header, you can have some additional header in some cases, but the first part of it is mandatory.

Continued

And, after that the actual data that you want to send ok.

Now, let us see what are the fields that are there in the IP header?

The way I have shown this table is that you see these numbers on top, these are the bit numbers, you see it starts with 0, it ends with 31, which means in every row there are 32 bits, which means 4 bytes and there are 5 such columns you see 5 such columns.

So, the header is total 20 bytes, 4 bytes in each row, I am showing 1, 2, 3, 4, 5, 20 and in addition you can have some optional header fields also.

Now, here we shall be looking at the function of these different header fields, there is a version, which will tell you about the IP version, this is header length, service type, total length of the IP packet, some identification of this packet, some flags, fragment offset, time to live, I will explain these things, protocol, header checksum and these are IP addresses.

You see source and destination IP addresses are 32 bits long, they occupy entire 32 bits ok.

Now let us see this different header fields one by one what they are

Now let us see this different header fields one by one what they are.

First we will indicate a version which is a 4 bit field, which indicates which version of IP you are using.

Now, here I am assuming this IP version 4, but there are some newer version of IP, version 6 which is also there, which is already being used in many places.

We shall be talking about IP version 6 later, but normally for our normal computers which we use, there you use IP version 4.

So, this first 4 bits will contain the number 0 1 0 0 which is 4.

The next 4 bit indicates the header length, what is the size of the header?

So, here the size of the header is expressed in multiple of 32-bit words.

So, as you can see that the first five rows were there and there are some options.

Continued

So, if it is only 20 bits of header, this will contain 5, but it can go up to a maximum 15 because it is 4 bits, in 4 bits you can have 0 to 15.

So, default is 5, 0101 which means 5 into 32 which means 160 bits ok.

Then there is a field which is 16 bits long, this is total length, this indicates the total size of the datagram including the headers.

Now in 16 bits what can be the size, 2 to the power 16 is 65536 so, many bytes.

And, if you subtract the header suppose in the minimum the header can be 20.

So, you go minus 20, that many bytes can be there as part of the data, that can be maximum size of the data in IP.

This is what IP supports.

Then there are some other fields which some of these are used very rarely.

Continued

For example, service type, this is an 8 bit field which contains some additional information which allows a machine to tell, whether you are wanting some special kind of

service.

Like I mean whether you want to give some higher priority to your packets, well if you give some higher priority, then your packets will be routed first.

You will get faster performance, you will get more bandwidth like that, but not all routers support this feature.

So, this feature is again used very sparingly, normally you do not use this.

Now, there is another very interesting field, this is called time to live.

This is an 8 bit field; 8 bit field means you can have a maximum value of 2 to the power 8 minus 1, which is 255, this can be the maximum value.

What this field contains, you see, you think of a network.

These are all intermediate nodes, these are the routers let us say.

Continued

So, these IP packets, they will flow from one node to the other.

Now, normally what do you expect that, you will be given the destination address, normally you would expect that the packet will be forwarded in the right direction and will finally, reach the destination.

But, so many things can happen in the network over time, some link might go down, some networking node might get down, some network, the routing table may also get corrupt ok.

So, what might happen is that instead of getting forwarded in the right direction your packet might accidentally get forwarded in the wrong direction also.

Now, in the extreme case what may happen, your packet might go along indefinitely in a cyclic loop.

Like for example, your packet might go along in the cyclic loop indefinitely, it will never go out.

If due to some reason, the routing table entries have changed or they have got modified in such a way that the packet will get forwarded like this only; now, you do not want such a situation to happen.

Now this time to live is a field which is put in the header with an initial value.

Continued

Now how much initial value that depends on the network, you can set that value suppose 50.

Say every time an a packet is forwarded from one node to the other the value in that packet is decremented by one, it becomes 49, next time 48, 47, 46, like that, like that whenever the value reaches 0, the packet is discarded.

So that you do not allow this indefinite cycling to go on, this is time to live, how long this packet can live ok, this is that.

And, protocol is another field, this is also 8 bits, this will identify which higher layer protocol you are using.

Like TCP and UDP are the most important protocols, they have a code, 8 bit code, there are other protocols also.

In that case you will have that that particular protocol number in 8 bit which high level protocol is being used here.

Then of course, you have this source and destination IP addresses.

As I said, these IP addresses are 32 bit addresses ok.

Continued

Then, you have some other fields which we shall be discussing later, like identification, flags, fragment, offset.

These fields are used for fragmentation, whenever the IP layer requires to break a packet into smaller packets and again it may have to combine those smaller packets together into a single packet, these are called fragments, fragmentation and reassembly.

So, to support that these three fields are required, this we shall be talking about later, when you talk about fragmentation and reassembly ok.

And of course, we talked about the variable width options header, you can have some additional header fields you can add.

These additional fields are required in some cases where some routers may be having some special support, like one interesting thing may be something called source routing.

You see normally the way a packet will be routed, you leave it up to the IP layer in the routers ok.

You send a packet to the destination address and let the routers decide, the IP layers in the routers decide.

But some routers may be having a feature called source routing, where you as the transmitter as this source know that this is the best route.

Continued

And I want my packet to follow this route; that means, this source is specifying the route; that means, you to specify that route, you need a longer information to be specified.

So, this options field can be used to specify the sequence of IP addresses of routers, which has to be followed to go to the destination, this is called source routing.

So, these options are there.

And, here lastly now for error checking, there is a field which is there, which is called header checksum.

Like, whenever a packet is transmitted, like let us say I transmit a packet, you receive that packet.

There may be a some error in transmission, some bits might get corrupt, 1 might get 0, 0 might become 1.

So, immediately whenever a node receives a packet, there has to be some simple checking whether there has been any error in transmission or not.

And, this header checksum is used for that purpose, the idea is like this; suppose you have the header and you have one field in the header, this is the checksum, let us say this is the checksum.

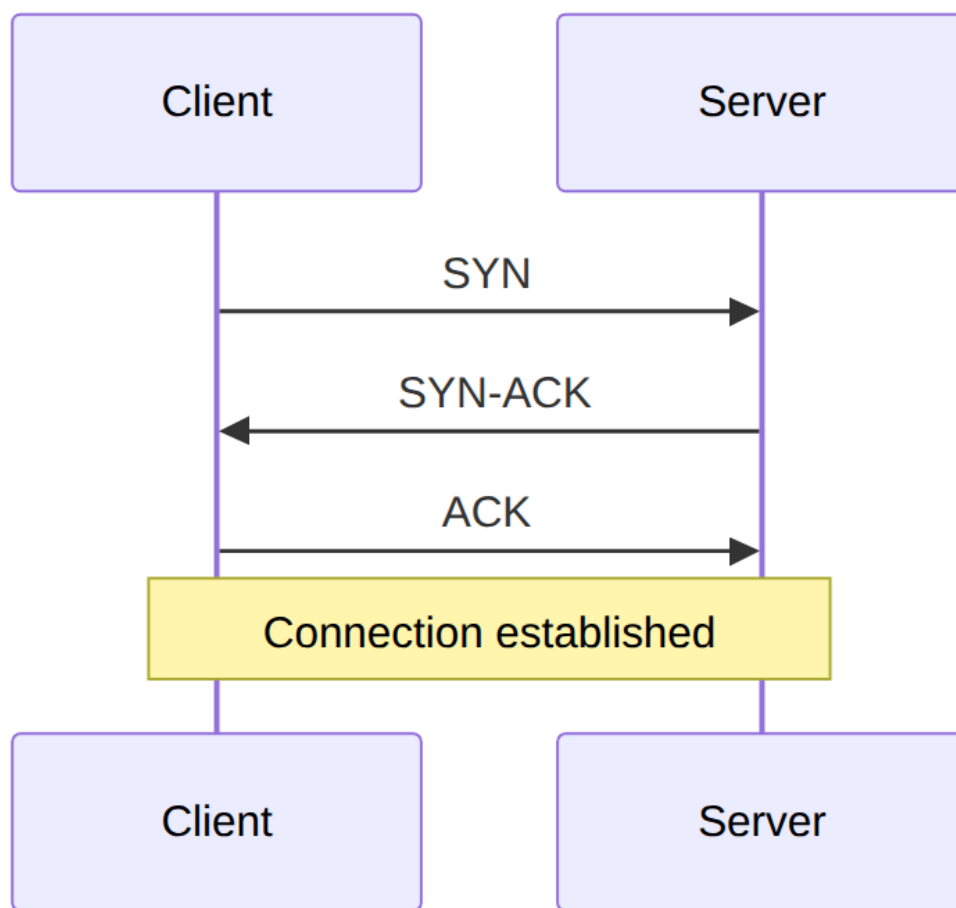


Figure 4: Architecture diagram

Tools mentioned

- Wireshark

Key terms

Term	Meaning in this lecture
So, default is 5, 0101 which	5 into 32 which means 160 bits ok.
Now, there is another very interesting field, this	time to live.
This is an 8 bit field; 8 bit field	you can have a maximum value of 2 to the power 8 minus 1, which is 255, this can be the maximum value.
Protocol is the, UDP is the 17, the value 17	UDP, this is the header checksum, which has been verified to be correct source address, IP, destination you see, means all these fields of the IP header you can view here if you want ok.

Exam-oriented recap

- So, in that sense the IP layer does not maintain or need not maintain any kind of history and it must be provided with sufficient information, so that it can forward the packet correctly to the final destination.
- Like for example, your packet might go along in the cyclic loop indefinitely, it will never go out.
- Say every time an a packet is forwarded from one node to the other the value in that packet is decremented by one, it becomes 49, next time 48, 47, 46, like that, like that whenever the value reaches 0, the packet is discarded.
- Like TCP and UDP are the most important protocols, they have a code, 8 bit code, there are other protocols also.
- Like, whenever a packet is transmitted, like let us say I transmit a packet, you receive that packet.
- So, immediately whenever a node receives a packet, there has to be some simple checking whether there has been any error in transmission or not.
- So, it is actually whenever a packet is received maybe by a computer, maybe by a router, this checksum computation is done typically automatically by the hardware.
- So, whenever a packet is received, the header fields are compared, the checksum is computed and the computer checksum is compared with the received checksum, whether they are matching or not.

L06: IP Addressing and Routing — Part I

Transcript: [06-ip-addressing-and-routing-part-i.md](#)

Video: <https://www.youtube.com/watch?v=IlRAeNleWUQ>

Week / theme: Week 2 — IP addressing, routing, TCP/UDP, subnets

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- So, let us continue with our discussion.
- Now, in this lecture I shall be broadly talking about IP packet fragmentation; IP addressing I will be talking about a little later.
- Firstly, we will be talking about IP packet fragmentation and there are broadly two types of fragmentation which is there, transparent and non transparent; we shall be looking at those.
- Now, this will be our network model; we will be assuming that we have a collection of networks, these N1, N2, N3, N4, these are different networks and you can see this small R's, this R's are nothing but routers.
- So, let us first look at transparent fragmentation; transparent means networks does not understand that fragmentation has taken place, it is transparent let us say.

What this lecture teaches

Core concepts from the lecture

So, let us continue with our discussion.

In this lecture we shall be talking about IP addressing and Routing, because as I mentioned that in the Internet TCP/IP plays a very important role and it is the IP layer which is at the network layer, which is responsible for all the routing of the packets and some addressing issues.

So, the title of the lecture is IP addressing and Routing part I.

Now, in this lecture I shall be broadly talking about IP packet fragmentation; IP addressing I will be talking about a little later.

Firstly, we will be talking about IP packet fragmentation and there are broadly two types of fragmentation which is there, transparent and non transparent; we shall be looking at those.

This fragmentation comes from the point that I had mentioned earlier, that when a packet is being given to the IP layer for routing, the packet may be too large, there may be some networks where the IP software or the layer is so configured that the maximum size of a packet is limited.

So, if the incoming packet is larger than that, then the packet may have to be broken up or fragmented into smaller packets ok.

This is the basic idea behind fragmentation.

Continued

So, let us come to this fragmentation as I said, we require because we need to break up a packet at times.

Now, the way it works if you look at the layering, you have the IP layer sitting at the network level and below the IP layer you have the data link or the physical layer or

the hardware layer whatever you call, let us call it the data link layer.

So, whenever the IP layer wants to send some packet for delivery, it will first be giving that packet to the data link layer ok.

So, the IP layer simply injects a packet into the data link layer and IP does not consider at all regarding reliability.

It is a basic datagram service where each of the packets are being sent out as independent entities and as I mentioned earlier, the packet might get lost.

Duplicate packets may get generated and the ordering of the packets is also not maintained.

There is no guarantee right.

Now, the point that I was mentioning, that at the layer of IP; in fact, at every layer there is a maximum size of the data unit that can be handled, that is primarily because of the size of the buffers, the number of bits that is reserved in the packet format and so on ok.

Continued

And this limit or limitation is referred to as Maximum Transfer Unit or MTU; the MTU can vary from one network to another.

So, when a packet traverses through multiple networks, it may encounter various different MTU values alright.

So, as I had said, suppose I have a large packet and it is trying to flow through a network, but MTU is rather small.

So, what will happen?

There will be a process called fragmentation which will take place, but the packet will be broken up into smaller packets and somewhere later the smaller packets will have to be put together again to get back the original packet, that part is called reassembly; so, fragmentation and reassembly.

And each of these fragments that are created they are treated as separate IP packets and are transmitted separately and this fragmentation is typically done by the routers when it receives a packet from some other network and wants to route it inside the network or maybe to the next network right.

This fragmentation broadly when you are doing fragmentation and reassembly, the process can be transparent or non-transparent, this we shall be seeing in some detail.

Now, this will be our network model; we will be assuming that we have a collection of networks, these N1, N2, N3, N4, these are different networks and you can see this small R's, this R's are nothing but routers.

Continued

This R's refer to routers; routers are essentially, they are networking devices which operate at the IP layer level ok.

So, whatever we are talking about regarding IP fragmentation, it will be taken care off inside that router ok.

So, you see there are routers at the boundary of the different networks and the different networks are typically connected through these routers.

For example, N1 and N2 are connected via these two routers, N2 and N2 and N4 are connected by these two routers and so on right.

Now, suppose there are also some host, some computers connected to the network, let us say I have one computer, a host H here and there is another host here, suppose I want to transmit some data packet from this host to this host in N4.

So, there can be multiple paths that the packet can take.

So, I am showing one of the path shown by this red dotted line where it will first be coming to this R, then this router to the N2, through N2 it will cross, then it will entered this and finally, it will reach the final network.

So, the idea that I was talking about may be N1 and N4 is able to handle large packets, large MTU, but N2 is an intermediate network through which the packet is flowing.

Continued

So, if N2 does not allow large enough packets, then there may be fragmentation and that fragmentation will typically be carried out by this router.

Now, this packet whenever it enters N2, this router will be fragmenting the packet into multiple smaller packets and then it will be forwarding ok, this is the basic idea.

So, let us first look at transparent fragmentation; transparent means networks does not understand that fragmentation has taken place, it is transparent let us say.

The idea here, whenever there is fragmentation, this subsequent network does not realize that fragmentation has taken place.

So, each of the fragments when they come, they are treated as independent IP packets and they are routed separately.

So, these intermediate networks are not at all worried about the reassembly part, only it is doing the fragmentation ok.

But, you see the idea is something like this, suppose I have a network here, a packet is coming, it is getting fragmented, it is coming out.

So, the subsequent networks does not need to know that a fragmentation has taken place.

Continued

The idea is that if there is fragmentation, then their exit router in the same network will be responsible for reassembly, it will again put together the fragments back.

So, that the original packet which was there, here it will again come out of this network.

So, fragmentation and reassembly happens inside this network and it is totally transparent to the other networks right.

So, the idea it follows is that say, suppose a large packet reaches a router which breaks it up into smaller packets or fragments and each of these fragments are routed and there is a constraint here, they must be sent to the same exit router; let us say RE, because in a particular network, there can be a multiple routers connected, there can be a multiple routers and connection to the outside world.

So, all the fragments must be sent to the same exit router, so that the exit router should be able to do the reassembly, put the fragments back together again ok.

This RE will reassemble the fragments into the original packet before it can forward to the next network, this is the basic idea, this is called transparent, because these subsequent networks are not aware of the fact that fragmentation is taking place.

Now, in general because a packet may be traversed in multiple networks on its way to the destination, this kind of fragmentation and reassembly may happen multiple times.

So, this is something you also need to remember.

Continued

So, talking about transparent fragmentation, I am illustrating this here with the help of this simple example.

Suppose I have a packet here, a larger packet which is being generated by this host in network N1, sent to this particular router, this router forwards it to this router of network N2.

So, this original packet reaches R.

Now, R sees that this packet is too large and the MTU of N2 is smaller.

So, what it does, it breaks up the packet into three smaller packets, these are the fragments and they are sent to the same exit router, there can be multiple other routers also, here I have not shown, I am showing only one router.

They are sent to the same exit router which will be again reassembling them back to the original packet and that packet will now reach network N4.

Now this particular router and network N4 will not be aware of the fact that there was a fragmentation earlier, because what it receives is the original packet, this is transparent fragmentation.

Now here one of the drawbacks for transparent fragmentation is, here we are saying that all packets must go through the same exit router, because the exit router will have to reassemble the fragments, but the problem is even if there are multiple exit routers all the fragments will have to go to the same router for handling.

Continued

So, that router might get overloaded.

So, if there was a scope for parallel processing using other routers, maybe the forwarding of the packets would have been faster, but here because we are using the same exit router, the burden on that router increases, it may become slow ok.

Now, another thing is that, talking about the exit router; exit router must be able to know that whether it had received all the fragments of a packet, so that it can put together all the fragments again.

There are two alternatives you can think off?

Now, each packet can come with a count field; count field will indicate how many remaining fragments are there.

So, when it reaches the exit router, exit router looks at the count field in the packet.

So, when the count field reaches 0 let us say, so, it will know that all fragments have come, now I can reassemble them or there can be a special delimiter like end of packet.

The last fragment of the packet will be marked specially so that the exit router will know that this is the last fragment, there are no more fragments after this.

Continued

So, now, I can assemble fine.

Now, now here as I said here, lot of overheads are encountered, there are two things, one is we are putting more burden on one particular router.

And secondly, after fragmentation the exit router must be doing a reassembly.

So, if this packet flows through multiple networks, there will be multiple possible fragmentations multiple reassembly again fragmentation again reassembly.

So, the amount of overhead is much higher here.

Now, coming to non-transparent fragmentation, this is what the IP protocol follows.

Here the fragmentation is not transparent to the subsequent networks, the subsequent network will know that fragmentation has taken place.

Suppose, a router divides a packet into four fragments; so all those four fragments will go to the next network as separate packets.

Continued

So, the subsequent networks will be receiving all these four smaller packets as independent separate packets ok.

So, there is no reassembly that is done in the intermediate stages.

So, the basic concept is that the fragments are not reassembled at any of the intermediate routers, fragments are created and let them flow to the destination through the intermediate networks or routers wherever they are.

Each fragment is treated as an independent packet as it said and the responsibility of the reassembly in this fragment lies with the final destination host.

Suppose, I am sending the packet to your computer finally, all the fragments will come to the IP layer of your computer and that is the IP layer of your computer where these fragments will get assembled again ok, this is how non transparent fragmentation works and as it said, the IP protocol in TCP/IP uses or follows non transparent fragmentation.

Now, non-transparent fragmentation, let me again also illustrate with an example.

Suppose this particular host in network N1 is generating a packet here, which has to be transmitted.

Let us say this packet is large enough.

Continued

So, this is broken up into two fragments P1 and P2, it depends on the MTU of N1; N1 is small.

So, it has to be broken into two fragments.

Now, suppose datagram can flow through any path; let us say P1 goes through N2 and P2 goes through this network N3 it may so happen.

Now, when P1 reaches N2 maybe the MTU of N2 is even smaller.

So, this P1 is broken up into three smaller fragments, let us say 1, 2, 3 and on the other hand P2 is broken up, let us say into two fragments 4 and 5 ok.

So, this 4 and 5 will be forwarded to this network, this N4 via some router and 1, 2, 3 will be forwarded to network N4 via some other router.

Now, as it said IP does not guarantee order of delivery of the packets.

So, maybe 1, 2, 3 will be received in some other order 3, 1, 2 or 2, 1, 3 in that order and may be here in the order of 5 4.

Continued

Finally, all these five smaller packets or fragments will reach the destination host and destination host will be looking at some of the fields in the header and will be putting together the fragments in proper order and it will generate the original packet.

This is how non-transparent fragmentation works.

Fragmentation can be taken, can be done by the different routers, assembly or re-assembly will be done only at the final destination host, this is the basic idea.

The advantages you can clearly see, here you can use multiple exit routers for transmitting the fragments and therefore, you can have better utilization of the networking resources the links, this will result in higher throughput quite naturally because you are using multiple paths together.

But drawback is that as the previous example shows, so, if the original packet is large, there will be multiple fragments.

In the previous example, there are five fragments created and each of these five fragments are IP packets.

So, there will be about 20 bytes of header ok.

So, this header overhead will be increasing.

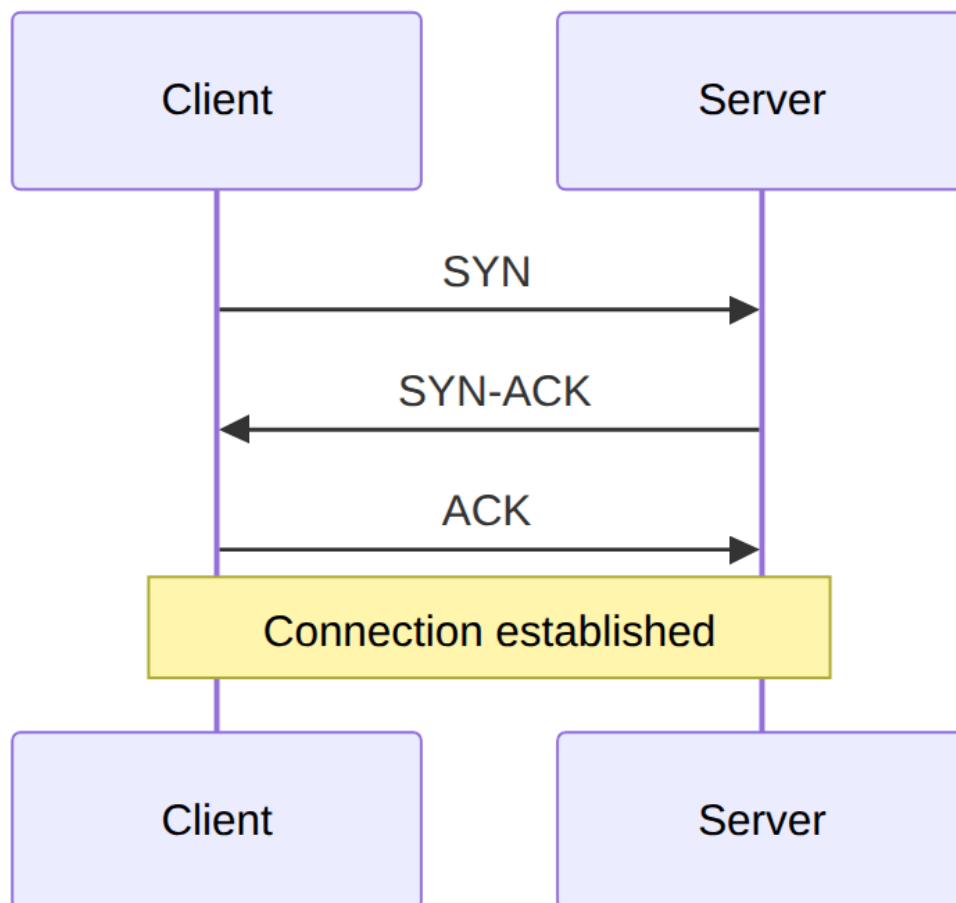


Figure 5: Architecture diagram

Key terms

Term	Meaning in this lecture
If M equal to 1, it	there are more fragments coming after this, but if it is 0, it means that this is the last fragment of the original packet, there are no more after this ok.
M equal to 0	there are no further fragment, this is only one single piece, single fragment packet.

Exam-oriented recap

- So, whenever the IP layer wants to send some packet for delivery, it will first be giving that packet to the data link layer ok.
- Now, this packet whenever it enters N2, this router will be fragmenting the packet into multiple smaller packets and then it will be forwarding ok, this is the basic idea.
- The idea here, whenever there is fragmentation, this subsequent network does not realize that fragmentation has taken place.
- So, all the fragments must be sent to the same exit router, so that the exit router should be able to do the reassembly, put the fragments back together again ok.
- So, this is something you also need to remember.
- Now, another thing is that, talking about the exit router; exit router must be able to know that whether it had received all the fragments of a packet, so that it can put together all the fragments again.
- And secondly, after fragmentation the exit router must be doing a reassembly.
- And this 13-bit field actually this specifies the offset in multiple of 8 bytes, you should remember this.

L07: IP Addressing and Routing — Part II

Transcript: [07-ip-addressing-and-routing-part-ii.md](#)

Video: <https://www.youtube.com/watch?v=GF0cAU5p6KU>

Week / theme: Week 2 — IP addressing, routing, TCP/UDP, subnets

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- So, let us continue with our discussion on IP addressing and Routing.
- Now, in this lecture, in the part II of it, here we shall be talking about some aspects of IP addressing and the concept of IP address classes.
- Well, when you talk about IP addressing we need to understand what is the basic role of IP.

What this lecture teaches

Core concepts from the lecture

So, let us continue with our discussion on IP addressing and Routing.

If you recall, in our last lecture, we talked about how packet fragmentation and re-assembly happens in IP.

Now, in this lecture, in the part II of it, here we shall be talking about some aspects of IP addressing and the concept of IP address classes.

Well, when you talk about IP addressing we need to understand what is the basic role of IP.

IP is a protocol of the TCP/IP family that works at the network layer level, and IP ensures some kind of uniqueness of each node or computers that are connected to the Internet.

At the level of IP, we assign some kind of address to every computer or host and that address is supposed to be unique, because it is very clear, if it is not unique, if you are sending a packet to some other computer, and if there are multiple computers with the same IP address, there will be confusion, the routers will not know where to forward and therefore, this uniqueness in address is very important ok.

So, each host connected to the internet, it is very important that it has to be identified by unique IP address.

And the way IP address is defined, it is a 32-bit quantity.

Continued

So, you can clearly see, you can have 2 to the power 32, so many different unique addresses which is about 4 billion.

You can have 4 billion such IP addresses that are unique right.

Now, here I will take an example, this 32-bit number or this address, this is a little difficult to remember 0 1 0 1 1 0 1 1, 32 binary digits.

So, to make it little concise, there is something called a dotted decimal notation where you divide the 32-bits into 4 bytes which are called octets; and each of this 8 bit quantities, we express in decimal.

And you write them as the decimal numbers, W X Y Z separated by dots.

This is called the so called dotted decimal notation.

Now, we shall see that this 32-bit network address that we are talking about, there are two parts in this address.

The first part identifies a network, there are many networks in the world; Internet is essentially a network of networks.

Continued

So, the first part will identify which network I am talking about.

And the second part identifies a host.

This is somewhat similar to the way we specify the address of our house, we specify our country, we specify a city, we specify a street, then we specify a house number.

Similarly, here we specify a network, and within the network which computer a host, which host ok.

So, depending on the way this partitioning is carried out, we can define these IP address classes, this we shall see.

Now, the dotted decimal notation, let us take an example here.

Suppose, here we have 32-bit IP address, you see there are 32-bits, this we have divided up into 4 8-bit chunks, octets; and each of these 4-bits chunks, we are expressing in equivalent decimal ok.

And this 32-bit number, we are expressing in a concise form like this 66 dot 134 dot 48 dot 126.

Continued

This is so called dotted decimal notation which is much easier to write, express and also remember ok.

This is the basic idea.

Now, this hierarchical addressing, I have already talked about that the way we address a computer in two parts, address of the network and address of the host.

So, here I am just repeating it again.

So, every host or computer that is connected to the Internet is identified by two things, a two tuple, first one I mentioned already the network number.

Now, network number has to be unique, all the networks in the world must be assigned a unique number, each of them must be unique.

So, this has to be done by some central authority.

So, there has to be some central authority which manages this network numbers.

Continued

And whenever you require to set up a new network, they will give you a unique number, this is the idea.

And the second part is the host number within that network ok.

This of course, can be managed by the local authority.

Suppose, in your organization, you have a unique network number, but inside you have 1000 computers, you can number them 1, 2, 3, 4, 5, 6 up to 1000 that is up to your local administrator to manage that numbering ok.

So, the host numbering can be done by the local network administrator.

And now when the routers forward, route the packets, they do not look at the host part, they will only look at the network number part, because the first task of the router will be to send the packet to the correct network.

So, only the network number is looked at and once the packet reaches the correct, the correct network, the router of that network will receive that packet and inside the network it will forward it to the correct host.

This is how routing of the packet occurs.

Continued

Now, talking about the IP address classes, there are five defined IP address classes, where class A, B, C are used to assign addresses to individual computers.

They are called unicast addresses.

unicast means address of a single node.

Class D is used for multicast; I want to send a packet to multiple computers at the same time, I can use class D address.

And class E is reserved, you can use it for some special purpose if you want.

Now, which class a particular IP address belongs to.

Suppose, I give you an IP address like this, 32-bit IP address.

Just by looking at the first few bits, you will be able to identify which class the address belongs to ok; we will see how.

Continued

And we shall also see later there are some special purpose IP addresses which serve some or which have some special meaning ok.

And this kind of IP addressing where we define the classes is sometimes also referred to as the classful model of addressing fine.

Let us look at the class A address first.

In class A address, the way we identify that this is a class A addresses, that it must start with 0, any address that starts with 0 is identified as a class A address.

The next 7 bits identify the network, the last 24-bits identify the host within a network, because there are seven network bits that can be 2 to the power 7 combinations.

Well, out of them there is one special, all 0 is not used it, taking away the all zero combination there can be 127 possibilities, there can be 127 class A networks.

And inside each such network there are 24-bits in the host part, 2 to the power 24 combinations are possible, out of that two of the combinations are used for some special purpose, we will see later the all 0 and all 1 combinations.

So, if you take away these two, this become 2 to the power 24 minus 2 which is of the order of 16 million.

Continued

So, this is used for very large networks up to 16 million computers in a single network right.

And if you look at the address range, the first bit is 0, and you can have all 0.

So, 0 dot 0 dot 0 dot 0 and 0 followed by all 1, it comes to 127 dot 255 255.

This is in dotted decimal notation.

So, just by looking at the address, you can know that if it is in this range, this will be a class A address ok.

Now, let us look at class B; class B is uniquely identified again by the first few bits, by the first 2 bits.

Anything starts with 0 means class A; anything that starts with 1 0 means class B ok.

Now, here we have 14-bits for your network, 2-bits are left a side and 16-bits for the host.

Continued

So, you can have again similarly 2 to the power 14 minus 1 which is about 16000 networks.

And each network and have 2 to the power 16 minus 2 about 65,000 computers.

This is how class B networks are.

And if you again try to look at 1 0, starting with 1 0 and remaining all others can be all 0 to all 1s.

So, the range will be 128, see 1 0 and all 0, the first byte will be 128, all are 0 0 0 up to 191, 1 0 followed by 6, this is 191, 191 255 255 255.

This is the range of class B addresses.

Similarly, let us look at class C.

Well, for class C, again anything starting with 0 is class A; starting with 1 0 is class B; starting with 11 0 is class C.

Continued

So, you see there is some kind of uniqueness, just by looking at the first few bits, you will be able to identify which address class it is ok.

Now, in class C, you see there are 21-bits you are using for networks and only 8-bits for the host, that means, networks where relatively fewer number of computers are there for such things, for such cases class C is most suitable.

With 21-bits you can have about 2 million networks, and with 8 bits in the host, you can have 254 computers, hosts per network.

And again if you just expand 1 1 0 followed by all 0 to all 1, you can find this is the range of the IP addresses.

So, just if you are given an IP address, just if you remember this numbers, by looking at the IP address, you will be able to know which address class this IP address belongs to.

Now, class D address starts with similarly 1 1 1 0, this is a multicast address as I said.

So, if, so the network if you are sending a packet using this multicast address starting with 1 1 1 0, then the packet will be delivered to all the computers in that network, the idea is something like that.

I am not going into detail.

Continued

The address range similarly is like this.

Now, address distribution is like this, you see class A networks are few in number, but in each network the number of computers can be huge.

So, in terms of the total number of address, this class A occupies the maximum, followed by class B and finally, followed by class C ok.

This is the address distribution if you are interested to know, this is how it works.

Now, I mentioned that there are some special purpose IP addresses also which are used, which are reserved for some special purposes.

First there are something called private IP addresses.

Suppose, I have an organization; I want to use some computers inside my organization to communicate among themselves.

I did not want or I did not need to have unique IP addresses which are not used anywhere in the world, because I am using only within my organization and boundary inside.

Continued

So, I can use something called private IP addresses which can be used by others also that is not supposed to be used, used publicly with other networks, only inside your network you can use this private addresses.

Like for class A, this is identified as a private address, any address that started 10 dot, 10 dot something that is a private class A IP address.

Class B there are actually 16, 172 dot 16 to 172 dot 31, there are 16 such private class B networks which are identified as private.

Similarly, for class C, this 192 dot 168, this is a private class C network.

So, there are many cases while you use private addresses inside your organization, but when you are going out of the organization, you have use some kind of a network address translator or some kind of an address translation mechanism to get a unique address or a proxy server, you can access the outside world, this is how you work ok, all right.

Now, there is something loopback or local address, any address that starts with 127 that is assumed to be a local address, local address means, it never goes out of the network, it will remain inside, that is referred to as a local address ok.

Suppose, even from your computer if you try to send a packet to 127 dot something, it will come back to your own computer, it will not go to anywhere else, that is referred to as a loopback, the local address.

Default network, any network address is 0 0 0 0, that is your default network.

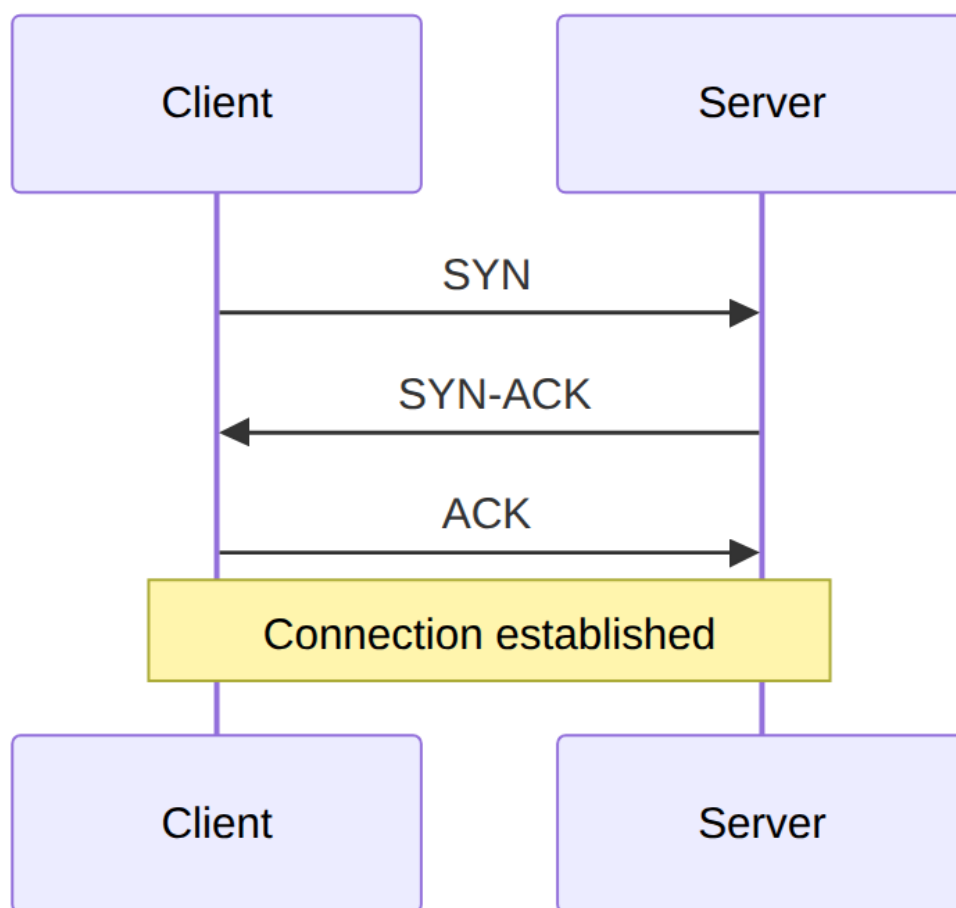


Figure 6: Architecture diagram

Key terms

Term	Meaning in this lecture
This unicast	the so called dotted decimal notation. address of a single node.
Anything starts with 0	class A; anything that starts with 1 0 means class B ok.
And similarly if we use an address all 1s, all 255, it	limited broadcast, means broadcast within your present network.
And if you write all one like this, this	the broadcast address of the network.

Exam-oriented recap

- So, each host connected to the internet, it is very important that it has to be identified by unique IP address.
- Now, here I will take an example, this 32-bit number or this address, this is a little difficult to remember 0 1 0 1 1 0 1 1, 32 binary digits.
- This is so called dotted decimal notation which is much easier to write, express and also remember ok.
- Now, network number has to be unique, all the networks in the world must be assigned a unique number, each of them must be unique.
- And whenever you require to set up a new network, they will give you a unique number, this is the idea.
- In class A address, the way we identify that this is a class A addresses, that it must start with 0, any address that starts with 0 is identified as a class A address.
- So, just if you are given an IP address, just if you remember this numbers, by looking at the IP address, you will be able to know which address class this IP address belongs to.
- Now, in the next lectures, we shall be looking into some more detail on the TCP and UDP protocols, because these are the two very important protocols in the TCP/IP protocol suite that runs on top of the IP layer.

L08: TCP and UDP — Part I

Transcript: [08-tcp-and-udp-part-i.md](#)

Video: <https://www.youtube.com/watch?v=0RFYDNG6IU4>

Week / theme: Week 2 — IP addressing, routing, TCP/UDP, subnets

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- In this lecture, we start with some specific discussion on the TCP and the UDP protocols, what they are, and how they work in the context of the TCP/IP protocol suite.
- These are the few things we shall be covering in this lecture.
- Let us now look at specifically what are the roles of TCP, what does the TCP protocol does.
- There are no errors, everything that I sent will be received correctly by the other end, but it is actually the TCP layer which handles all errors and tries to recover from the errors ok.

What this lecture teaches

Core concepts from the lecture

In this lecture, we start with some specific discussion on the TCP and the UDP protocols, what they are, and how they work in the context of the TCP/IP protocol suite.

So, this is the first part of this lecture TCP and UDP.

In this lecture, we shall be talking about the basic functionalities and roles of TCP and UDP, in particular we shall be talking about something called port numbers and how they relate to data communication at the transport layer level, and how TCP and UDP uses port numbers ok.

These are the few things we shall be covering in this lecture.

Now, we have already talked about the TCP/IP protocol suite.

Now, in TCP/IP if you look at the transport layer level, at the transport layer level, there are two different protocols that are used, one is TCP which is the short form for transmission control protocol, other is UDP which is the short form for user datagram protocol.

Now, the basic idea is whenever there is a user program, some program, which is trying to send or receive data from some other host on the Internet, it interacts with either the TCP or the UDP layer.

Suppose, I want to send the data, I can give my data to TCP, TCP in turn will give it to IP and IP will actually send or route the data via the data link layer which is below it ok.

Continued

This is how it works.

The user processes, they will interact with either TCP or UDP and TCP and UDP will be using the IP layer which is below it at the network layer level.

Now, this diagram shows what I just now said diagrammatically.

So, this is where we are at the transport layer level.

So, at the transport layer level, we have two alternatives, either TCP or UDP.

So, whenever there has some user process which is running, there is a choice, it can either choose to use TCP or it can choose to use UDP.

It depends on the application, what kind of facilities are the transport layer level the user process wants and accordingly it can do that.

But the point to note is that you see whatever you use TCP or UDP, at the network layer level, you have no choice; you have this IP and only IP.

Continued

So, both TCP and UDP, they interact with the IP protocol at the network layer level which you recall is a datagram base service which is unreliable, datagrams might get lost, duplicates might get generated, order may not be maintained.

These things I mentioned a number of times, ok, fine.

Let us now look at specifically what are the roles of TCP, what does the TCP protocol does.

TCP provides a connection-oriented service, it is reliable.

Reliable means if there is any data corruption during transmission, some data packets are getting lost or gets corrupted, TCP is supposed to keep track of that.

And if something is wrong the source or the sender will be requested to again send the relevant part of the data, so that the receiver can receive the data in a correct way that is reliability.

Full-duplex means it is a two-way communication, a can send to b, b is also sending some data back to a.

So, it is not a one-way communication.

Continued

Byte-stream service means whenever some message is being transmitted, both the sender and receiver keeps track of how much data have been sent and how much data is yet to be send in terms of the number of bytes.

So, each byte has a number 1, 2, 3, 4, 5, 6, 7, 8, and both sender and the receiver keeps track of these byte numbers.

How many bytes I have already sent, how many bytes I have not yet sent and similarly for the receiver, it sends back sometime some acknowledgement, it says that well I have received all bytes up to byte number 1050, you can now send me the bytes after that, something like that ok.

So, to provide this end to end reliability that TCP provides, you see because TCP provides reliability, the applications that use TCP, it does not have to care about anything.

It assumes that my network is very reliable.

There are no errors, everything that I sent will be received correctly by the other end, but it is actually the TCP layer which handles all errors and tries to recover from the errors ok.

Now, TCP provides the reliability using several mechanisms.

There is a checksum where each packet is verified using a checksum mechanism whether there packet has become corrupted, whether there is an error in the packet or not.

Continued

Positive acknowledgement, the receiver sends back some acknowledgement to the sender that well I have received all bytes up to 1050, this is the positive acknowledgement.

So, now, the sender knows I have to send byte number 1051 onwards now, rest have been received correctly.

Timeouts, well if the sender sees that the acknowledgement is not coming from the receiver till some defined interval, there will be a timeout mechanism which will be enforced and the entire message or the segment will be resend.

It will assume that there was some network error, maybe the data did not receive the other end, let me send it again.

And end-to-end flow controls, so this speed of the sender and receiver may not be the same.

So, dynamically both the send and receive at the TCP layer adjusts the speed of data transmission and data receiver, recipient, receiving, sending and receiving, so that this speed mismatch is taken care of Well, now in addition to this, TCP also handles connection establishment and termination, because it is a connection-oriented service similar to virtual circuits we talked about earlier.

There is a concept of connection establishment and termination, but the thing to note is that you are talking about connection at the TCP layer level, but down below, it is IP.

Whatever you do at the TCP layer level ultimately IP will be sending out datagrams, they will be following different paths and they will be reaching the destination in any other order.

Continued

But the TCP layer gives that illusion to the user process that this is a connection-oriented service ok, this is how it works.

And TCP also you see because of the datagram at the IP layer, the data might reach the destination in some arbitrary order.

TCP will put together all received data in the correct order, which is called sequencing.

So, that the user process at the receiving end will get the data in the correct order always, that is one of the main responsibilities of TCP and this is what we mean by connection-oriented.

There is a virtual connection which is maintained and the user process assumes or I mean, it gets a feeling that there is a connection and all data are coming through that connection in the same order ok.

But actually TCP has to work quite hard in order to have all these features incorporated.

Well, in contrast UDP is very much similar to IP, it does not do much beyond that, it does not try to ensure any kind of reliability.

It provides a connectionless kind of a service, it is a datagram service.

Continued

TCP is a connection-oriented that is why, some connection establishment and connection termination are required.

But UDP does not require any connection establishment, each packet is being sent independently.

And because it is like a datagram service, this is also unreliable just like IP ok.

There I just two additional things which are there in UDP which IP does not have, at the transport layer level it incorporates a checksum to verify whether the UDP packet is correct or has been corrupted.

If there is some corruption there will be a checksum error possibly.

And there is something called port numbers which are also added.

Port numbers actually identify the user processes that are interacting with UDP or TCP, we will talk about this.

Yes, now you see you think of a computer, you think of a particular computer on the Internet.

Continued

There can be several user programs which are running P1, P2, P3, many user programs are running on this computer.

And they may all be trying to communicate some data over the network, over the network it is sending and receiving some data ok.

Suppose, P1 has send some data to somebody and that somebody is again sending back a response, a packet is coming back.

So, when the packet reaches this computer, see how will this computer know that this packet has to be sent to P1 and not P2.

There has to be something in this packet, in the header which must clearly identify that this packet has to be delivered to P1, and that is this port number.

On each computer every running process is identified with an unique port number.

And whenever at the transport layer level a message has been transmitted, either we using TCP or UDP in the header, there is a port number and that port number specifies which process is sending that message, and at the other end which process is supposed to receive that message.

Those port numbers identify that; that is the role of port number mechanism, to uniquely identify the data packets associated with each process.

Continued

So, each process will have a unique port number for every connection it is maintaining ok.

So, as this diagram shows, diagrammatically we will list the same thing.

Suppose on a particular computer, there are three running processes.

And when they are communicating over the Internet, over the network, they will be assigned some port numbers, let us say port number 10, 20 and 30.

So, when process 1 is sending some data packet; when we sending out some data packet, it will specify the source port number which is 10.

And when a response comes back, in the response the destination port number will be set as 10, so that this computer will know that 10 is what, 10 is this process 1, this packet has to be delivered to process 1.

So, this is how the transport layer in this computer will take care of sending and receiving of the packets and delivering them correctly to the appropriate application or process ok, this is how it works.

Now, in TCP and UDP, this port numbers are 16 bit quantities; well, 16 bit is quite large, you can have 2 to the power 16 which is about 64000, 64 k.

Continued

You can have that many active connections, active processes running and this quiet large.

And you see, there are certain applications which are well known, they are the uniquely identifies by some port numbers, like you think of your mail, SMTP, SMTP will be having some particular port number.

You think of some network application like telnet, telnet will be having some unique port number.

And if you think of any other kind of network application the common ones will be having some unique port numbers, so that suppose I am trying to send a mail, I know

that is my mail server, I always know that my mail server will be working on port number 25 and so, port number 25 or something whatever.

So, whenever I want to send a packet to my mail server that particular process, I will be sending a packet mentioning destination port number is equal to 25.

So, it will always go to the mail server.

Similarly, when I want to send some request to a web server, may be the web server is, is working on port number 80, I will be sending a packet saying destination port number 80.

So, my packet will automatically go to that web server, and web server processes it, and sends the request back to me.

Continued

So, by this unique port numbers, I can uniquely identify and contact a particular process which is running on the other end right.

Just one thing I will tell you, well in an internet scenario, if you think of a situation like this, this is a sender, this is a receiver.

Well, often we talk about client-server scenario, let us say this is a client and this is a server.

The client will be sending some request to the server.

With respect to the request, this is the sender, this is the receiver.

Now, when it sends a request, the client is supposed to know which process at the other end, it is trying to contact to.

So, the destination port number must be already known to the client ok.

As I told you for all the common applications, the destination port number is well-defined, everyone knows about it.

Continued

This is the overall picture, again I am showing here the MAC address or the hardware address we already mentioned earlier, this is a 48-bit address, at the IP layer you have a 32-bit IP address.

Now, at the TCP or UDP level, you have a 16-bit port number or port address.

So, again just repeating physical address ensures uniqueness of the network interface cards, every network interface card you plug into your computer that will be having a unique 48-bit address.

At the IP address level, every computer you want to connect to the Internet must have a unique 32-bit IP address.

Every connection to the Internet, to the router, to the network has to have a unique IP address.

Port number or port address, every user process running on a computer must have a unique port address, these are the different levels of uniqueness that are maintained, so that process-to-process communication becomes very unambiguous and unique fine.

Talking about this port number, we already mention this briefly.

In a client-server scenario, I was talking about this is a client, who is sending request to a server, it is wanting some service.

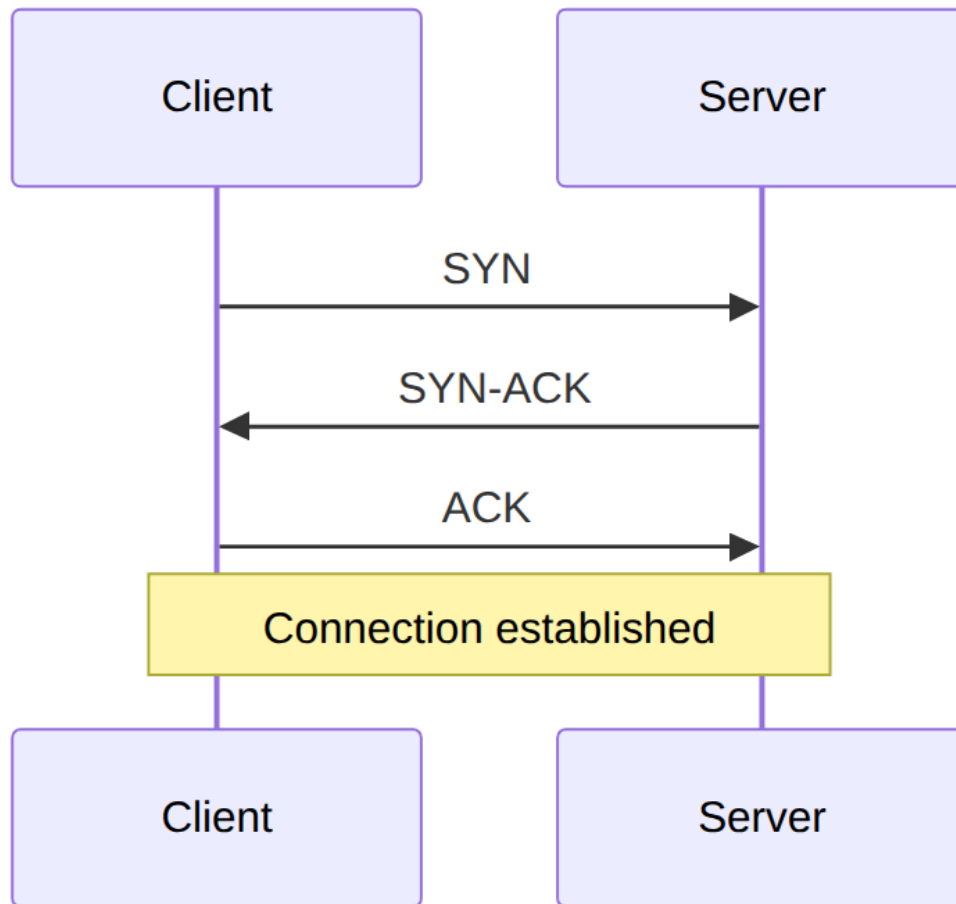


Figure 7: Architecture diagram

Key terms

Term	Meaning in this lecture
Reliable	if there is any data corruption during transmission, some data packets are getting lost or gets corrupted, TCP is supposed to keep track of that.

Term	Meaning in this lecture
Full-duplex	it is a two-way communication, a can send to b, b is also sending some data back to a.
Byte-stream service	whenever some message is being transmitted, both the sender and receiver keeps track of how much data have been sent and how much data is yet to be send in terms of the number of bytes.
It explains what that service actually	and there can be some aliases also, you can give some alternate names also.
But I do not have any unique port number	what I mean to say, I do not have any well defined port number, I am using it for sending a particular mail only.

Exam-oriented recap

- Now, the basic idea is whenever there is a user program, some program, which is trying to send or receive data from some other host on the Internet, it interacts with either the TCP or the UDP layer.
- So, whenever there has some user process which is running, there is a choice, it can either choose to use TCP or it can choose to use UDP.
- Byte-stream service means whenever some message is being transmitted, both the sender and receiver keeps track of how much data have been sent and how much data is yet to be send in terms of the number of bytes.
- So, that the user process at the receiving end will get the data in the correct order always, that is one of the main responsibilities of TCP and this is what we mean by connection-oriented.
- There has to be something in this packet, in the header which must clearly identify that this packet has to be delivered to P1, and that is this port number.
- So, whenever I want to send a packet to my mail server that particular process, I will be sending a packet mentioning destination port number is equal to 25.
- So, it will always go to the mail server.
- So, the destination port number must be already known to the client ok.

L09: TCP and UDP — Part II

Transcript: [09-tcp-and-udp-part-ii.md](#)

Video: <https://www.youtube.com/watch?v=EACepSXJx98>

Week / theme: Week 2 — IP addressing, routing, TCP/UDP, subnets

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- We continue with our discussion on the TCP and UDP protocols in this lecture as well.
- Now, you look at the whole TCP/IP protocol family and you can see that TCP sits here, TCP is here.
- Now, let us look at these diagrams.

What this lecture teaches

Core concepts from the lecture

We continue with our discussion on the TCP and UDP protocols in this lecture as well.

Here in the second part of this lecture, we shall be looking at the actual header fields of both the TCP and UDP protocols.

And, in particular we shall be looking at the process of TCP connection establishment and also connection termination.

Because, these are a few very interesting things which can lend itself to some kind of network based attacks that we shall be talking about ok.

So, let us start with TCP.

TCP we have already mentioned that at the transport layer level, we have a host to host connection facility.

So, TCP supports host to host communication facility and it supports a set of features.

The features are like these 2 processes can communicate among themselves, a process on a local machine and a process on a remote machine.

Continued

There can be a process here, there can be a process here, they can communicate.

Stream delivery service, the data that have been transmitted, these are called segments, they are considered to be as a stream of bytes, a stream of bytes are sent, a stream of bytes are received at the other end, this is said to be a stream oriented service.

Full duplex means both way communication, two way communication.

Multiplexing and demultiplexing means multiple active connections may be present between the two ends and multiple packets for multiple connections can flow through the same links and same logical links, these are called multiplexing and de-multiplexing.

And TCP ensures connection oriented and reliable service, this briefly we have mentioned earlier, that there are a few features using error control retransmission.

So, it tries to ensure this kind of features.

Now, you look at the whole TCP/IP protocol family and you can see that TCP sits here, TCP is here.

Below TCP, there are several protocols at the network layer level, where IP is the dominant protocol.

Continued

And, there are a few sister protocols which are there, which have very specific purposes.

So, the point to note, I am just repeating, so, although TCP provides a connection oriented reliable service to the applications at the higher layer, but all data transmission has to take place via IP and IP is a unreliable datagram service.

So, TCP has to do a lot of bookkeeping management, buffering, so that all this error correction and streaming and these reliable facilities can be provided to the application.

Now, let us look at these diagrams.

This first diagram actually shows you the conceptual view of TCP.

This is a sending process; this is a receiving process, on two different computers.

This is the TCP layer on this and there is the TCP layer on this.

So, TCP provides a host to host logical connection, this is not a physical connection, a conceptual connection.

Continued

So, you have this pipe that has been shown here, this represents the logical connection.

And, over this logical connection, a stream of bytes are flowing, this is how a TCP communication system looks like.

Now, let us look into some more detail inside TCP which is shown in the next diagram, you see both the sender and the receiver they maintain some kind of a buffer.

Buffer at the sender side will contain the message that it is trying to send.

It is a stream of bytes, the receiver also will maintain the buffer whatever comes it will put in the buffer.

And, both the sides are keeping track of how many bytes have been sent, how many bytes have been received and so on and so forth.

Like you see in the sender process, there are some bytes marked in pink, which indicates the bytes which have been sent, already sent.

The bytes marked in gray indicates which have not yet been sent and the process when it is generating some data to be sent, it will be storing them into the buffer from the next available location onwards.

Continued

In this way the buffer at the sending side will get filled up.

And, on the receiving side similarly some data is being received, this pink one says that this data have been received, but not yet read by the receiving process and whatever comes over the network, they will get stored out here.

Starting from the next available location onwards and the receiving process will be reading the bytes one by one from this pink part right.

Now, TCP maintains this buffer automatically, you see due to some error in between it may see that somewhere in between there is a vacancy, all bytes have not yet been received.

So, there will be an acknowledgment mechanism, the receiver will be requesting the sender to send back the missing bytes again, so that the receiving process can be forwarded the bytes in the correct order ok.

And, the other point we note is that which is just depicted in this diagram, a similar diagram.

It shows some basic data transfer unit called segments.

You see at the network layer level, we talk about packets are being transferred.

Continued

At the data link layer level, we talk about frames; frames are being transferred, but at the transport layer level, at the TCP level we talk about segments, data segments, which are nothing, but stream of bytes, multiple segments are being transmitted one after the other, this is the concept.

Now, let us look at the header format of a of the TCP protocol.

Just like IP, I am showing this bit numbers on top starting from 0 up to 31, which indicates that each row of this indicates a 32 bit quantity, 4 bytes.

So, first you have source port number, destination port number 16 bits each.

Then, we have a sequence number, well sequence number means I told you that data are being sent in units of bytes, stream of bytes, sequence number is like a byte number which byte number is being sent.

This is a 32 bit quantity.

So, which means maximum segment size can be 2 to the power 32 quite large.

Similarly, acknowledgement number is also 32 bits, acknowledgement number is coming from the other side, it tells the receiver, tells the sender that well I have received up to byte number so, and so, that is the acknowledgement number.

Continued

So, that the sender will know that well up to this byte number is already being received so, I need not keep it in my buffer anymore, let me remove it from my own buffer ok.

Then, there are some other fields like, there is a header length, just like TCP, there some reserved field, there are some flags, we shall talk about, there is a window, well window is like that buffer I told you, you know some data are being send, some data are being received.

So, there is some windows that are maintained by both sender and receiver, this is called a sliding window protocol, but I am not going into detail of this.

So, this window specifies this so called sliding window and is used for flow control.

Flow control means the maximum speed of data transmission you can control by changing the size of this window ok.

And of course, there is a checksum for correcting errors, detecting errors and there is an urgent pointer, sometimes as part of the TCP packet, you can send some urgent message, an urgent pointer actually points to some part of the data where that urgent message is located and some option field is also there.

So, the TCP header is typically minimum 20 bytes and can go up to 60 bytes.

So, a brief explanation, source port already I have mentioned, destination port, this identifies the process at the local end and at the remote end.

Continued

Sequence number, I am just repeating, this is used for keeping track of the bytes being transmitted for the purpose of reliability.

Each byte is assigned a 32-bit number and as the bytes are being transmitted that number is incremented by 1, 1, 1, like that fine.

Similarly, from the other side acknowledgement number is coming, the remote host will acknowledge receipt of the data; it will contain the number of the next byte expected.

Like say, suppose the receiver has correctly received up to byte number 1000, let us say.

So, it will be sending back an acknowledgment saying acknowledgment 1001.

So, it will be sending back an acknowledgment packet with the acknowledgement number field as 1001 ok.

Now, header length is a 4 bit field, it just similar to the IP and the IP header also has a header length field, it specifies the header length in multiple of 32 bit words ok.

So, minimum value is 5, because minimum header size can be 32, can be 20 bytes.

Continued

So, it can be 5 minimum.

Well, these are interesting, there are several flags, in the flag field, in TCP there are in fact, 6 flags URG, SYN, ACK, FIN for finish, RST and PSH.

They are used in combination for various reasons.

The 6 of the most important reasons are mentioned here, well if there is any urgent data, that is carried by the TCP, urgent data may be some kind of interrupt processing, some important thing need to be handled.

So, then the urgent flag will be set to one and I mentioned the urgent pointer, there is an urgent pointer field, it will be pointing to some place in the data part that urgent data will be stored there.

So, if the URG pointer is set to 1, it means that the urgent pointer is now active.

In TCP whenever you are doing a connection establishment, you are requesting for a connection then you set this SYN flag to 1 and ACK flag to 0, this this combination means connection request.

Similarly, connection confirmation I will show the details later.

Continued

So, when a connection is confirmed then the SYN and ACK both are set to 1 right and during sending if the sender has no more data, it has finished sending the data, then the finish flag will be set to 1, so that the connection can be released.

So, when the connection has to be terminated, this FIN flag is set to 1.

And, sometimes you may need to reset a connection due to some problem, then RST bit is used for this purpose.

Some connection attempt, you do not want to accept that connection, then also you can send the RST bit, the connection will be rejected.

And, there is the last bit, PSH bit this indicates a push function, this can also be used to indicate end of message.

So, when a message is finished, you can set the PSH flag to 1, it will tell the receiver that there is no more bits in the message ok.

So, these are the flags and the ways they can be used.

Window is a 16 bit field; now this field actually tells you how many bytes you can send, beyond the bytes acknowledged like let me given example.

Continued

This is the sender, this is the receiver, sender is sending bytes one by one, suppose the receiver has sent back an acknowledgement.

It has said that well I have received all bytes up to 1000, next byte I am expecting is 1001 ok.

Now, for the sender let us say the window field is 100, let us take an example.

This means that the sender can now start sending byte numbers starting from 1001 up to 1100 without waiting for an acknowledgment.

This window size actually tells how many bytes the sender can send without getting the acknowledgment.

Once the window is full, finished, all 100 bytes have been sent, then the sender will be waiting an idle.

It will be now waiting for the acknowledgement to come back.

So, as you can see if the window size is larger, you can send more data and wait less and then wait for the acknowledgement to come, but if the link is slow then possibly window size will be made smaller.

Continued

So, by adjusting the size of the window, the rate of transmission of data can be controlled.

So, you can use something called flow control, using this window, and the window size is referred to as window advertisement.

As, I mentioned it can be increased or decreased.

And finally, there is a checksum; checksum is used for error correction and this checksum is applied not only for the header, but for the entire data segment.

Like in the TCP protocol, you have the header, you have the data and there is also something called a pseudo header.

This checksum is applied to the entire part of it.

What does the pseudo header contains?

Pseudo header borrows some of the fields from the underlying IP, that is not strictly part of the TCP header, but it borrows something from IP like, source IP address, destination IP address, which protocol, it is using TCP or UDP and what is the length size?

Key terms

Term	Meaning in this lecture
Full duplex	both way communication, two way communication.
Multiplexing and demultiplexing	multiple active connections may be present between the two ends and multiple packets for multiple connections can flow through the same links and same logical links, these are called multiplexing and de-multiplexing.

Term	Meaning in this lecture
Then, we have a sequence number, well sequence number	I told you that data are being sent in units of bytes, stream of bytes, sequence number is like a byte number which byte number is being sent.
So, which Flow control	maximum segment size can be 2 to the power 32 quite large. the maximum speed of data transmission you can control by changing the size of this window ok.
So, if the URG pointer is set to 1 , it	that the urgent pointer is now active.
This	that the sender can now start sending byte numbers starting from 1001 up to 1100 without waiting for an acknowledgment.
So, this is going in this direction SYN-ACK	SYN is also 1 , ACK is also 1 ; both these flags are 1 , this will be going in this direction.
That	connection requests has come, not yet fulfilled.

Exam-oriented recap

- Because, these are a few very interesting things which can lend itself to some kind of network based attacks that we shall be talking about ok.
- The 6 of the most important reasons are mentioned here, well if there is any urgent data, that is carried by the TCP, urgent data may be some kind of interrupt processing, some important thing need to be handled.
- In TCP whenever you are doing a connection establishment, you are requesting for a connection then you set this SYN flag to 1 and ACK flag to 0 , this this combination means connection request.
- Because, this information are also important so that TCP can verify that whether the packet that has come, the message that has come, is actually coming to the right IP address or not.
- Then it can assume that due to some error this connection is not going through, let me discard this, but the problem is, this feature of TCP, I means actually has led people to mount some kinds of attacks on systems.
- So, this is a simple attack scenario I am just mentioning, suppose some hackers decide that I have to target that particular computer.
- So, even the legitimate connections will get denied, this is something which is called denial-of-service attack.
- So, this was one of the simpler ways to mount denial-of-service attacks based on the TCP protocol right ok.

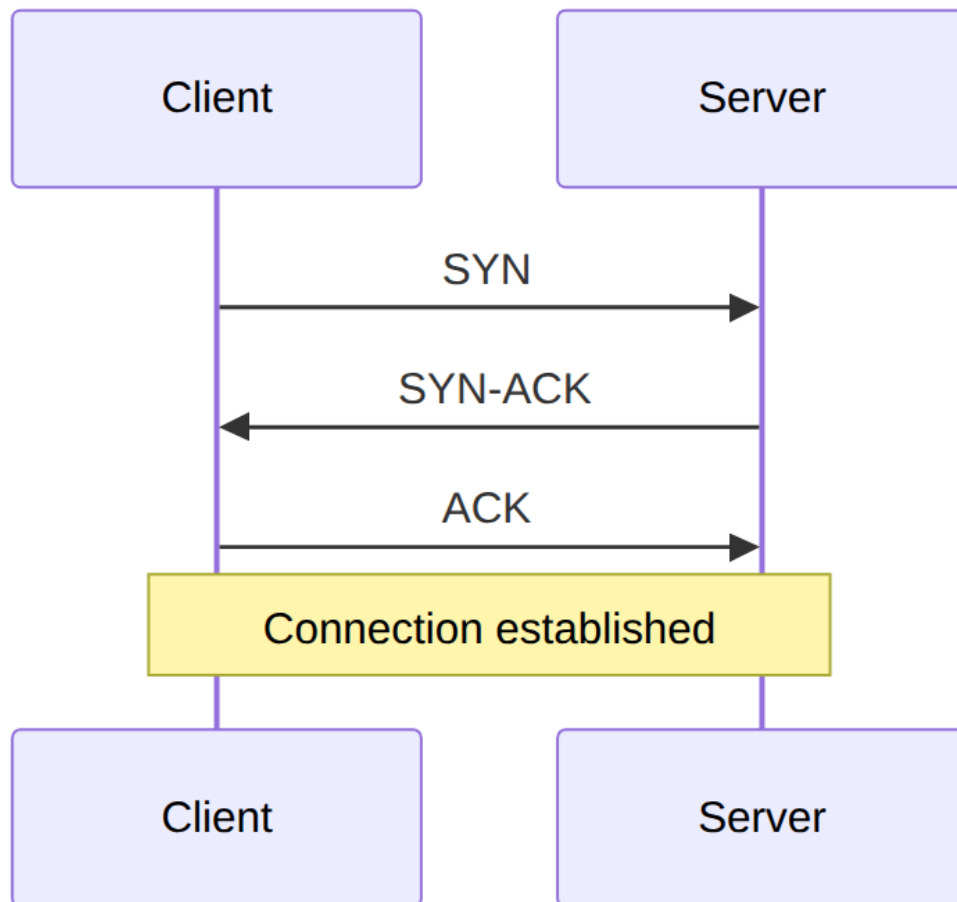


Figure 8: Architecture diagram

L10: IP Subnetting

Transcript: [10-ip-subnetting.md](#)

Video: <https://www.youtube.com/watch?v=-q8B75e9LOg>

Week / theme: Week 2 — IP addressing, routing, TCP/UDP, subnets

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- In this lecture, we shall be talking about IP sub networks or subnets.
- Now, let us look at the concept of network masks.
- Now, we have already seen, how this IP networks are; class A, class B, class C, we have talked about.
- Now, when you talk about subnets, the idea is a network, an IP network, it can be class A, B or C.
- We are further splitting into smaller sub networks; that is the concept of subnets.

What this lecture teaches

Core concepts from the lecture

In this lecture, we shall be talking about IP sub networks or subnets.

Now, we have already seen, how this IP networks are; class A, class B, class C, we have talked about.

Now, when you talk about subnets, the idea is a network, an IP network, it can be class A, B or C.

We are further splitting into smaller sub networks; that is the concept of subnets.

So, the topic of this lecture is IP Subnetting.

Now, here we shall be talking about these IP subnets and IP masks, how they are used and we shall be talking about two methods, which can be used for efficient creation of this kind of sub networks.

One is called Variable Length Subnet Mask or VLSM; other is Classless Internet Domain Routing or CIDR ok.

So, let us first start with IP subnet.

Continued

Now, earlier we have already talked about the class A, B or C IP networks.

Now, there we had seen, that we are using some kind of a two level hierarchical address to identify a particular host on the network.

The address consists of a network part and it consists of a host part.

The network part is classified as class A, B or C like that.

Now, when you are using IP subnets, we are introducing or we are adding a third level in the hierarchy.

We are using a network portion and host portion in addition, there is a sub network portion or subnet portion and the subnet portion is specified by using something called network masks.

This is what we are doing.

Now, let us look at the concept of network masks.

Continued

First, we talk about something called natural masks.

Natural mask means class A, B or C, in its purest form.

There are no sub networks; that is what is meant by natural mask.

Let us see what the natural mask is; when we take the example of a class A network, let us say the network addresses 10 dot 0 dot 0 dot 0, well you see conceptually what it means.

The first 8 bits indicates the network number, the remaining 24 bits indicates the host.

What I do, I define something called a mask.

Mask is a stream of 0s and 1s, well not arbitrary stream, first few bits will be 1s followed by all 0s.

1s indicate that these are bits corresponding to network; 0s indicate these are bits corresponding to host.

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So, when I say first 8 bit is network, last 24 bits is host, I write the mask like this, first 8 bits are 1s, remaining 24 bits are 0s.

So, when I write this mask again in dotted decimal notation, it becomes 255 dot 0 dot 0 dot 0.

So, class A network will be having a natural mask of 255 0 0 0, because the first 8 bits represent network.

So, first 8 bits are 1s.

So, let us take a specific example.

Suppose, we have a specific IP address of a host on this class A network sorry, it is a 10 0 0 20, this is the IP address.

This is my IP address and already we have seen that the natural mask is 255 0 0 0.

So, if I write it in binary 10 0 0 20 is this and 255 0 0 0 is this.

Continued

Now, this mask actually tells you, how many bits of the address we need to consider to know the network number, means if I do a bit by bit AND operation, bit by bit ANDing, first 8 bits are one in the mask, if you do AND so the first 8 bits will be 0 0 0 0 1 0 1 0 which is 10 in decimal, but remaining all bits are 0s, if you do an AND everything will be 0.

So, in dotted decimal notation, it will 10 dot 0 dot 0 dot 0, which will indicate the network number.

So, from the mask if you do a bit by bit AND, you are masking the host part, making it all 0 and you are extracting the network number, that is the main purpose of the mask.

So, natural mask for class A we have already seen, in a very similar way for class B, the natural mask will be first 16 bits, a network, 255, 255 then 0, 0, for class C first 24 bits indicate network 255, 255, 255 and then host, last 8 bits.

So, these are the natural masks.

So, natural mask has fixed division, but when we use so called subnets, we can have arbitrary masks, not necessarily this kind of subnet masks.

So, here as I have said when you are using subnetworks, here again, we are using masks and these masks are very flexible.

Like for example, in a class C network, I can have 254 hosts in a network, if I want I can divide it into four parts; wherein each part I can have let us say 50 computers each, 50, 50, 50, 50, this will be subnetworking.

Continued

Well, we shall see how we can do this.

Basically, what we will do in the IP address, as we know there are two parts; one is the network part, other is the host part.

What we are doing, we are taking the host part.

Now, in the host part we are again doing some kind of a partitioning, subnetwork and host.

Some of the bits of this host part we are using to identify the subnetwork number and the remaining bits to indicate a host within the identified subnetwork right.

So, let us take an example of; so, let us take class A network 10 dot 0 dot 0 dot 0, for which you recall the natural mask was 255 dot 0 dot 0 dot 0.

This was the natural mask, but now we are using a mask of 255 255 0 0.

In this network there were 4 parts, in this address there were four parts; first was the network, the remaining three was the host.

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Now, when I say 255 255 0 0, essentially I am saying my first 16 bits are my network, 255 255, but already the first 255 is network, because it starts with 10.

So, the next 255 also I am borrowing as the network, this will be my sub network.

So, as if this will be my network, subnetwork, host, host.

So, if I now have an address like this; for example, 10 dot 5 dot 0 dot 20 then, if we apply the natural mask of class A 255 0 0 0, you will get the network part 10 0 0 0.

Now, if we apply this subnet of mass 255 255 0 0, you will be getting 10 dot 5 dot 0 dot 0.

10 is already there, so the subnet of number will be 5 and whatever remains last part, last 16 bits is 20.

This is how sub networking works.

So, here I am just illustrating this.

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This is the IP address I was talking about, in binary I am writing like this 10 5 0 20, mask is this.

So, if I do a bit by bit AND, again I mean I will get 10 dot 5 dot 0 dot 0.

So, this will give you my sub network number.

Now, already because this is a class A network, I know that this is my network number.

So, whatever remains that will be my sub network, this 5.

So, essentially what we have done in a class A network.

Let us say 10 0 0 0, there were 24 bits for the host.

So, you had a maximum of 2 to the power 24 minus 2 hosts possible, but now we are using this second byte to identify a sub network.

Continued

So, you are left with only the last 16 bits.

In the last 16 bits how many hosts can be there 65534, 2 to the power 16 minus 2 and, because this can have 256 combinations, you can have up to 256 subnets and each of the subnets will be having up to 65534 hosts.

This is how sub networking works.

Now, here pictorially whatever we have said bit by bit ANDing, I am showing for default mask and also subnet mask.

Let us take an example here of a class B address.

This is a class B IP address starting with 144; you can check 144 starts with 10, which is class B.

So, for IP for class B address the default mask is 255 255 0 0.

So, if you do a bit by bit AND, you will be left with 144 16 0 0 which is your network address, but if you are using subnet masking, the same IP address, see 192 means what?

Continued

192 means this pattern, first two bits are 0, which means in a class B the last 16 bits indicated host, but in this subnet I am also using 2 more bits, these 2 bits I am taking out.

So, I am left with 14 bits for the host.

So, if I have one address like this and if I have a subnet mask like this, means if I do a bit by bit AND again, you can say this 144 16 will remain and 192 and 72 if you take a bit by bit AND, these bits were all become 0, only this one will remain 0 1 0 0 0 which is 64.

This will be the network address ok, network id including the sub network.

So, 144 16 is the class B network and 64 is the sub network number within that class B network.

This is how you can extract the sub network number from a network.

Now, we have something called variable length subnet mask.

The idea is that the same network depending on a scenario, you can configure with different masks.

Continued

Like earlier, what I said if I use the example, I had said just the previous example; I am splitting a particular network into certain number of subnets.

A number of hosts per subnet is fixed, but for variable length subnet mask I am saying, I can use variable like let us say I have a big network, I first divide it up into equal and one part I make half, this part I do not make half and one part I make 4, like this unequal pieces, I can make.

This is the concept of variable length subnet mask, let us see how it works and obviously, this will allow better utilization of available address space.

Well here, I am taking a very specific example suppose, I have a class C network 192 203 17, 24 bits are network and 0 last is the host part ok.

Let us say in an organization there are three departments; D1, D2, D3, I am trying to make three different sub networks and the number of computers are 110, 45 and 50.

Now, in a class C network, you can have 254 hosts total.

So, you have let us say a total class C network, I am having 254.
Now, if we divide it into half, it will be approximately 127 something.

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Now, in this you can accommodate D1, because it is 127, D1 requires 110, but in the other part you again have 127, but you need D2 and D3.

So, you can make the other part into half and this is enough for 45 and 50, you can put D2 in one and you can put D3 in the other.

So, the idea is something like this.

So, there are two subnet options we will see, the first one; let us say that the, if you can do without VLSM, by using normal subnet masks can we do?

Let us say we have a class C address just using some subnet mask with class C, 255 255 255 dot some value of X.

X can be either one followed by 7 0s or it can be 1 1, it can be so many things, 1 1 1, so many options.

So, let us explore whether any of these can satisfy our requirement.

So, I have shown a table for various values of X, if X is 128 means 1 followed by 0s, number of subnet is 2 with 128, 128 each approximately, if I make it 1 1, which is in decimal 192 sorry.

Key terms

Term	Meaning in this lecture
Natural mask means this pattern, first two bits are 0, which	class A, B or C, in its purest form. in a class B the last 16 bits indicated host, but in this subnet I am also using 2 more bits, these 2 bits I am taking out.
This 128 All 0s and 192	1 followed by eight 0s, 7. what?

Exam-oriented recap

- This is an added advantage here.
- Classless here, we have no concept of class A, B or C and this gives certain advantages, like the routing tables in the routers, they get reduced in sizes, because we do not have to store the whole masks like that.
- I can do any arbitrary partitioning, this is the advantage of CIDR.
- First rule is that in CIDR number of addresses in each block you are defining must be a power of 2, because you are using certain number of bits for address network, certain number of bits for host.

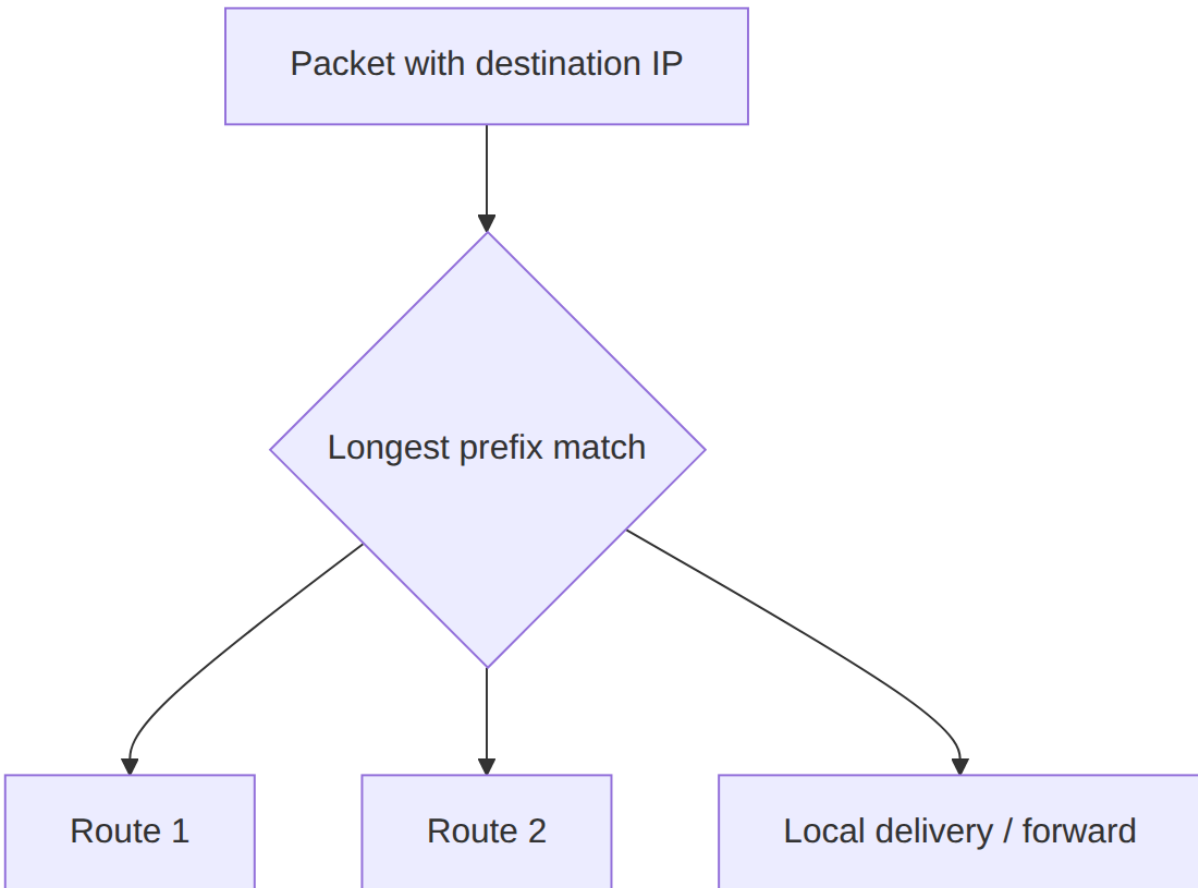


Figure 9: Architecture diagram

- Now, the beginning address in each block must be divisible by the number of addresses in this block.
 - So, there can be 16 hosts, 4 bits, what it says; the beginning address in each block must be divisible by the 16.
 - So, a block that contains 16 addresses can never have the starting address as this, because this is not divisible by 16, but you can have 64, because 64 is divisible by 16.
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